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Liu et al.

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(54) **FLIP-CHIP
SEMICONDUCTOR-ON-INSULATOR
TRANSISTOR LAYOUT**

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(51) **Int. Cl.**

H01L 23/528 (2006.01)

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H01L 23/482 (2006.01)

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H01L 23/522 (2006.01)

H01L 23/66 (2006.01)

H01L 25/07 (2006.01)

H03F 3/213 (2006.01)

(52) **U.S. Cl.**

CPC **H01L 23/5286** (2013.01); **H01L 23/4825**
(2013.01); **H01L 23/49838** (2013.01); **H01L**
23/5226 (2013.01); **H01L 23/66** (2013.01);
H01L 24/05 (2013.01); **H01L 24/13** (2013.01);
H01L 24/16 (2013.01); **H01L 25/072**
(2013.01); **H03F 3/213** (2013.01); **H01L**
2223/6616 (2013.01); **H01L 2223/6644**
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None

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Primary Examiner — Hung K Vu

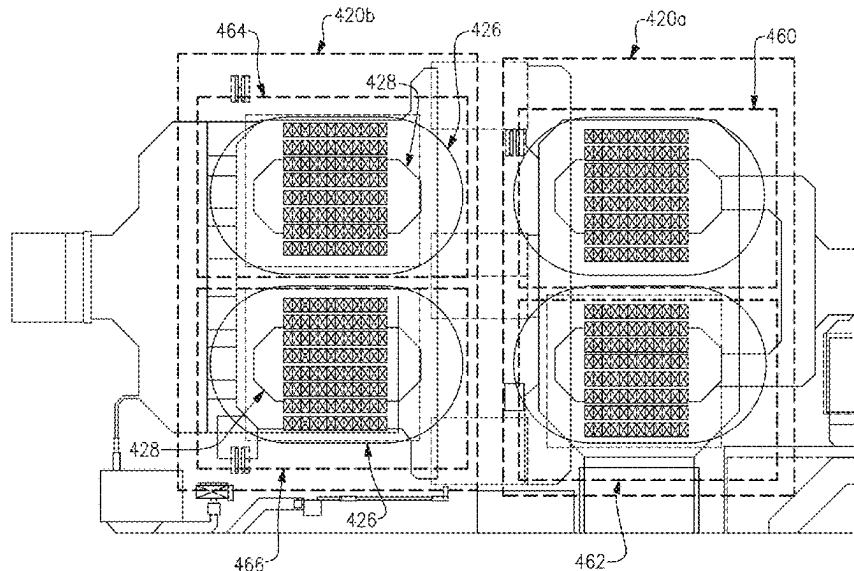
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ABSTRACT

A flip-chip semiconductor-on-insulator die includes a sub-
strate layer, an active layer, an insulator layer between the
substrate layer and the active layer, a first metal layer, and
a first via layer between the active layer and the first metal
layer. The die at least first and second contact pads and a
transistor including a first terminal formed within the active
layer. A first portion of the first terminal falls within a
footprint of the first contact pad and a second portion of the
first terminal falls within a footprint of the second contact
pad.

20 Claims, 25 Drawing Sheets



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- (60) Provisional application No. 62/957,733, filed on Jan. 6, 2020, provisional application No. 62/957,229, filed on Jan. 5, 2020, provisional application No. 62/957,097, filed on Jan. 3, 2020.

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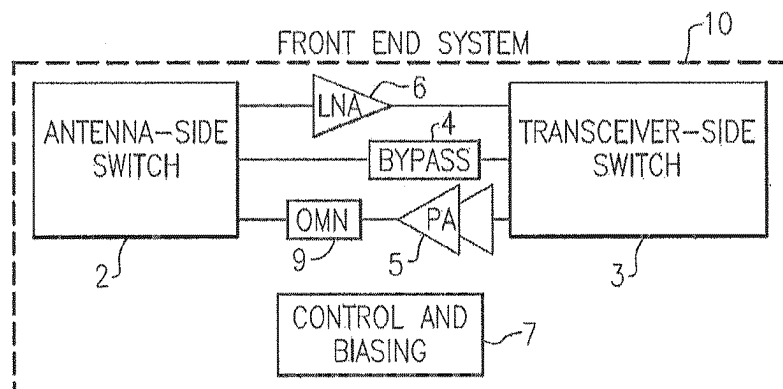


FIG.1A

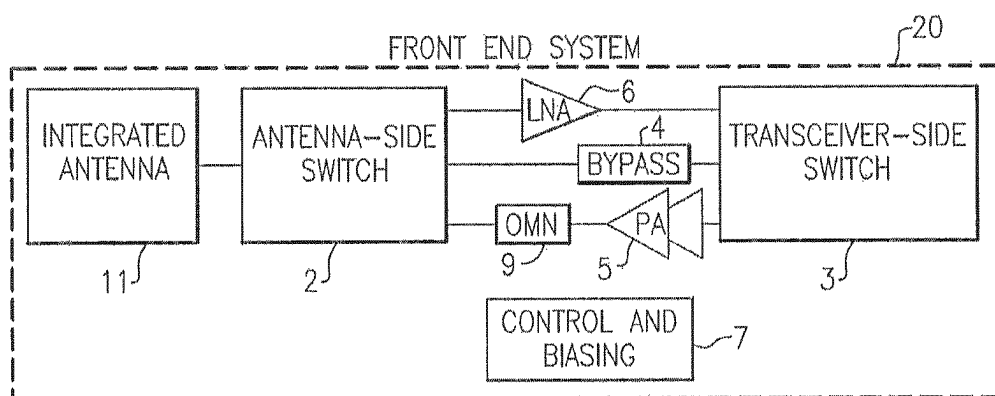


FIG.1B

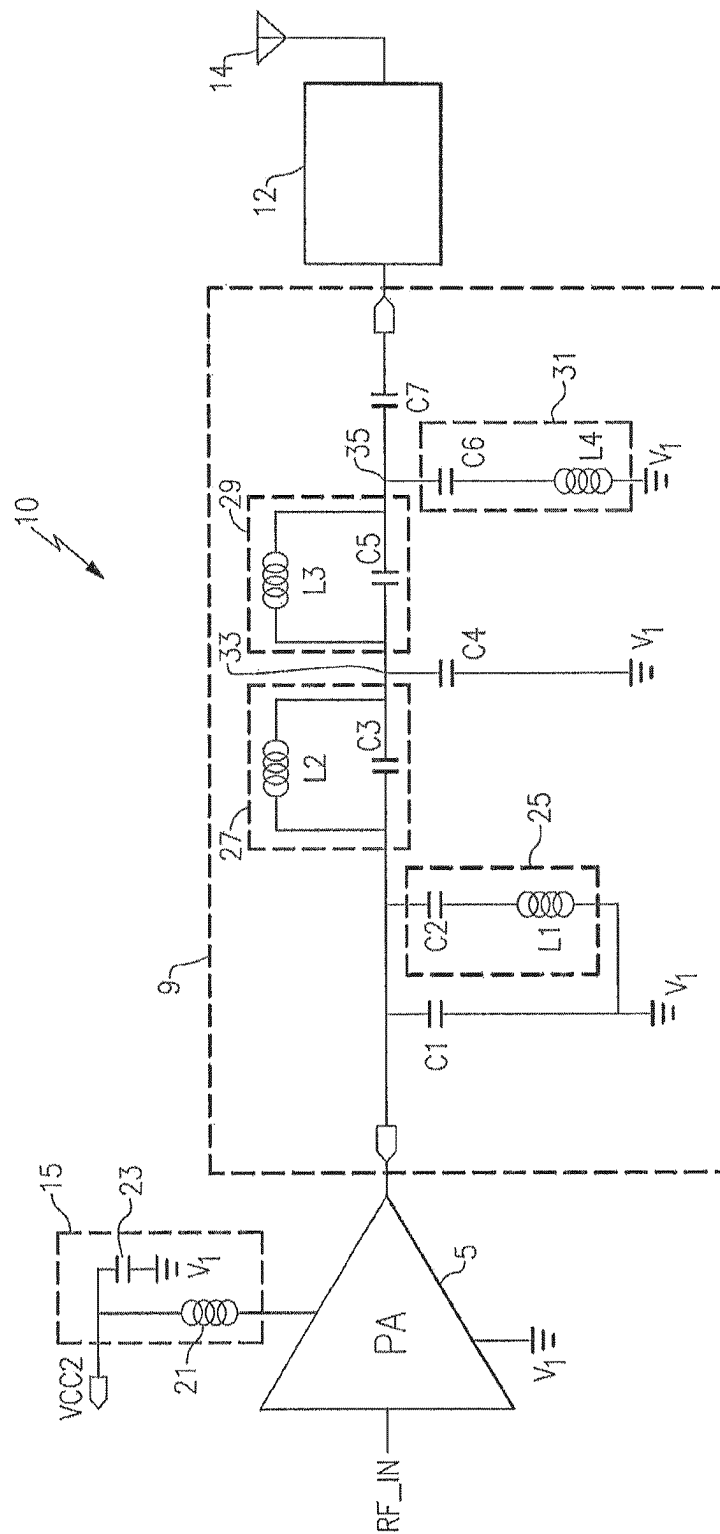


FIG. 2

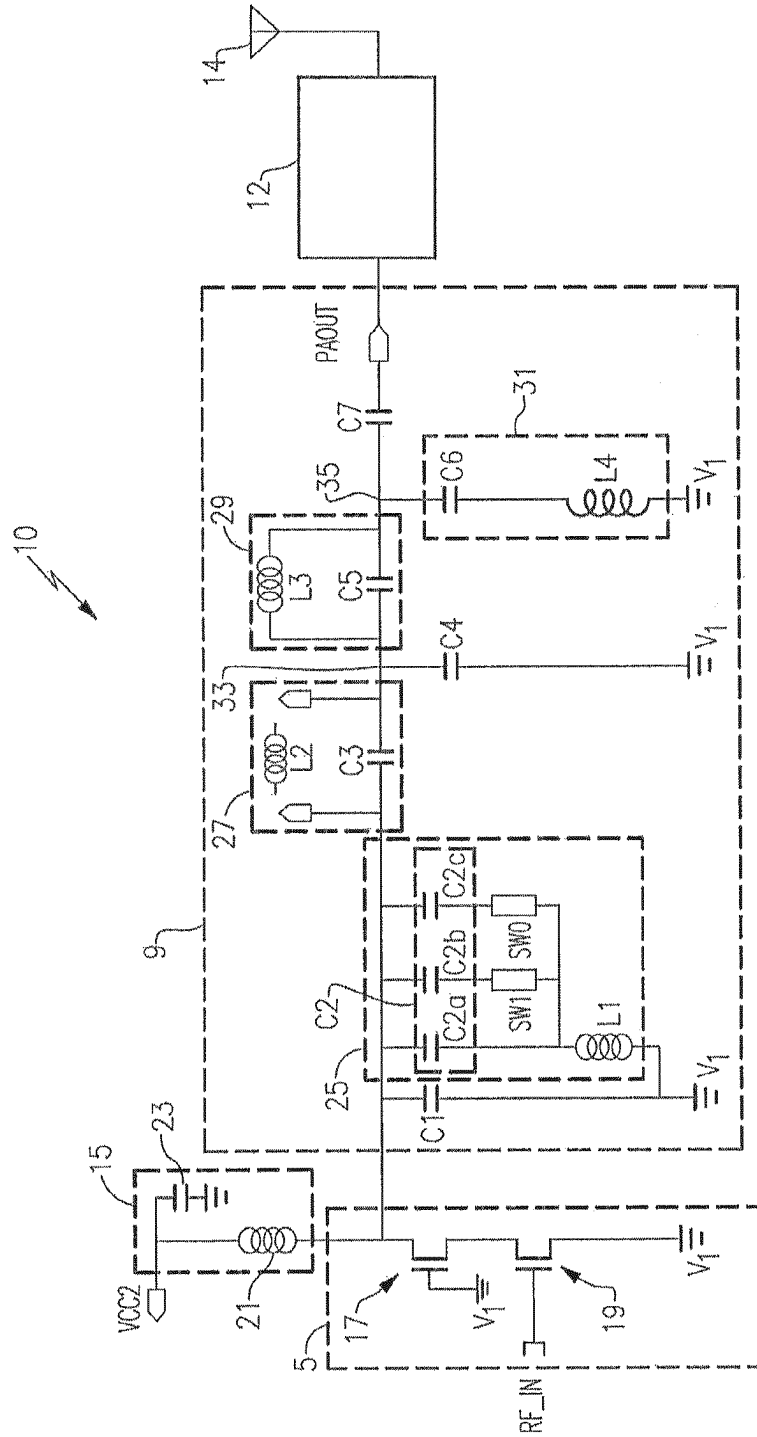


FIG.3

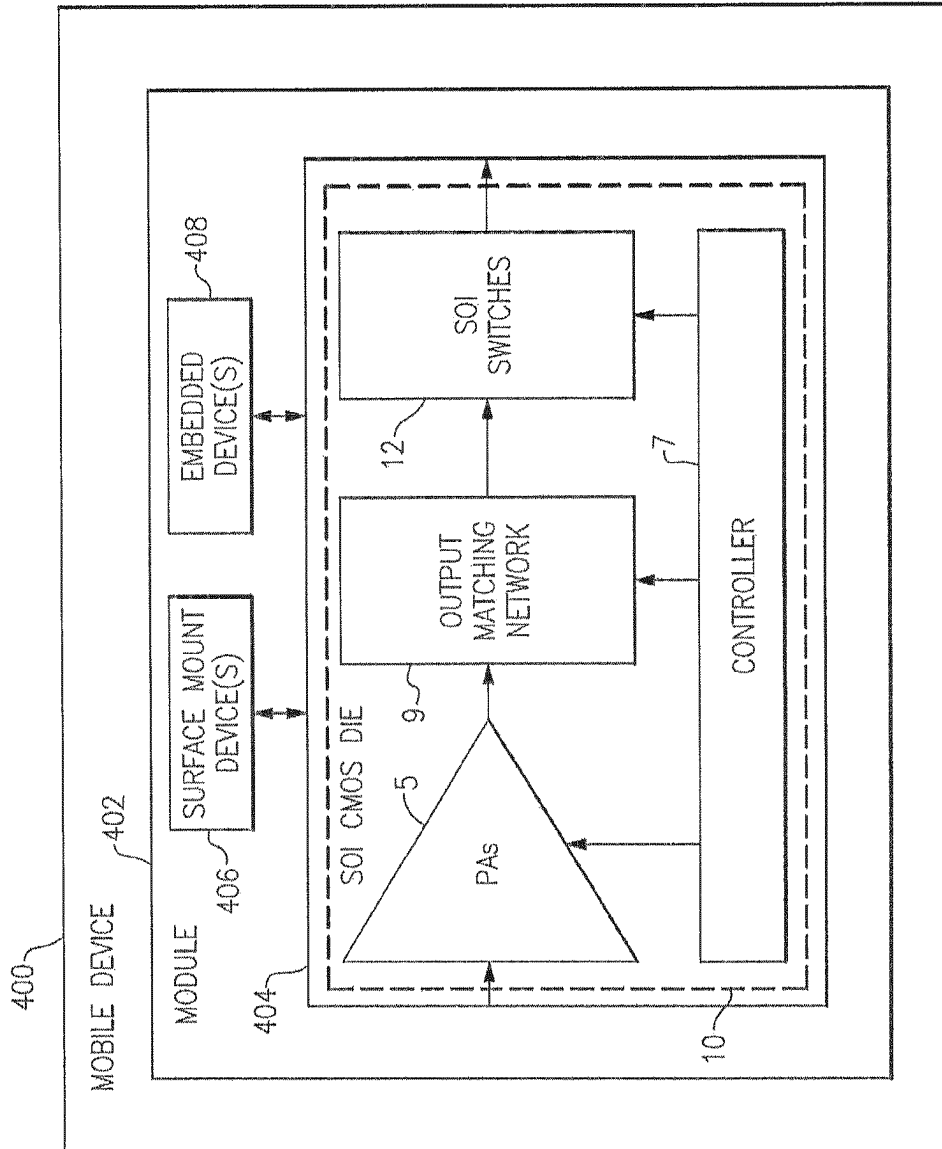


FIG. 4

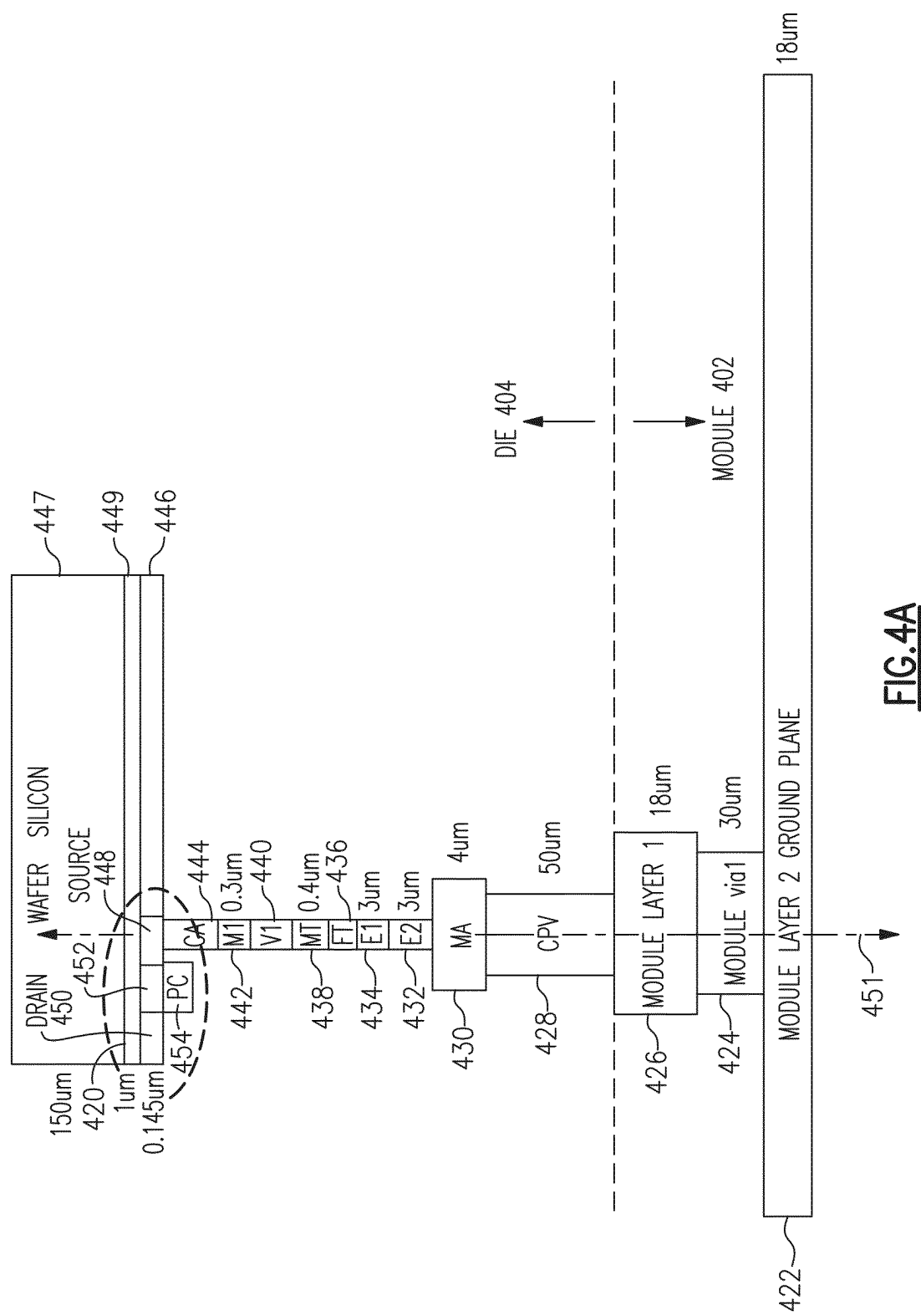


FIG. 4A

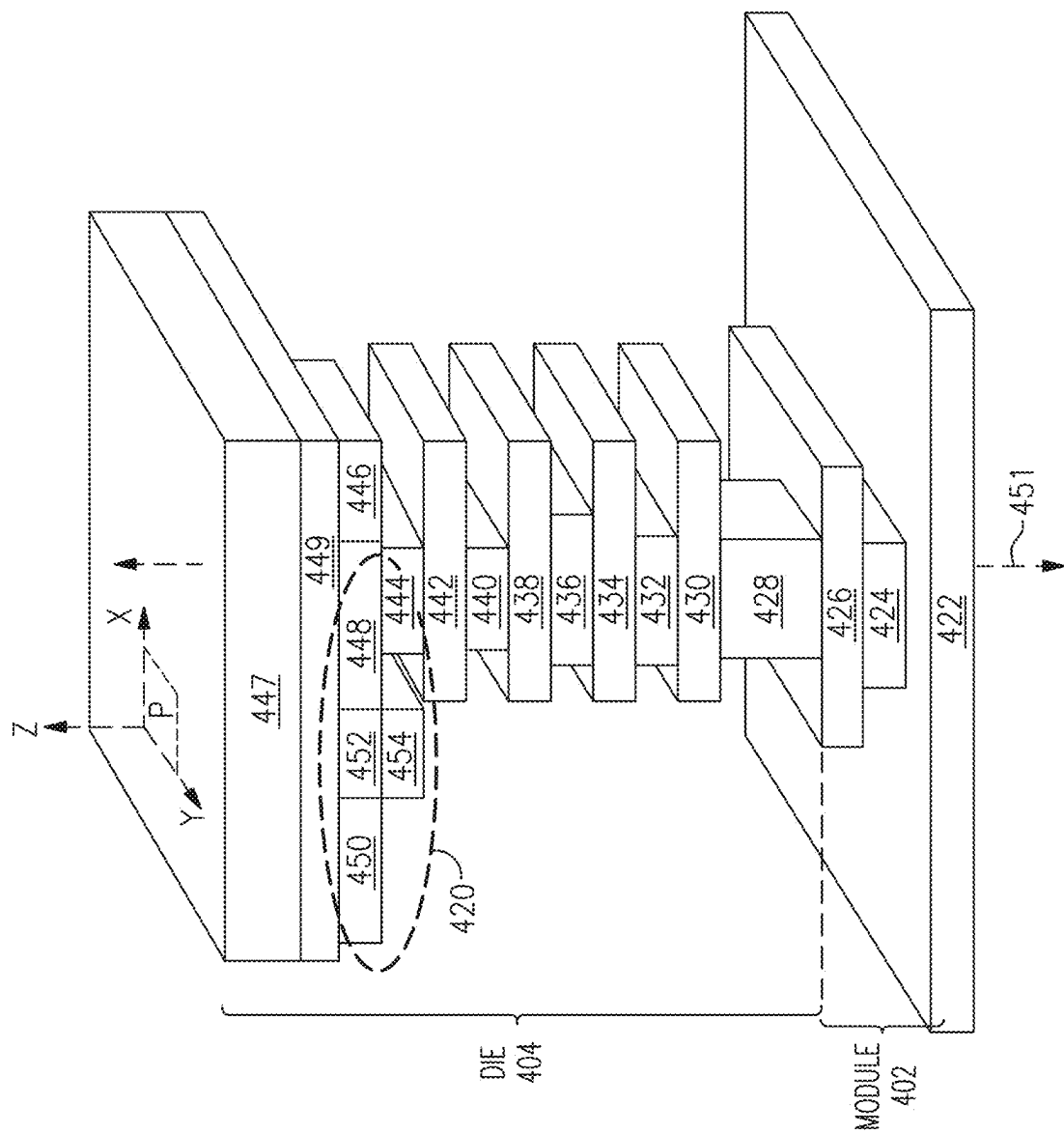
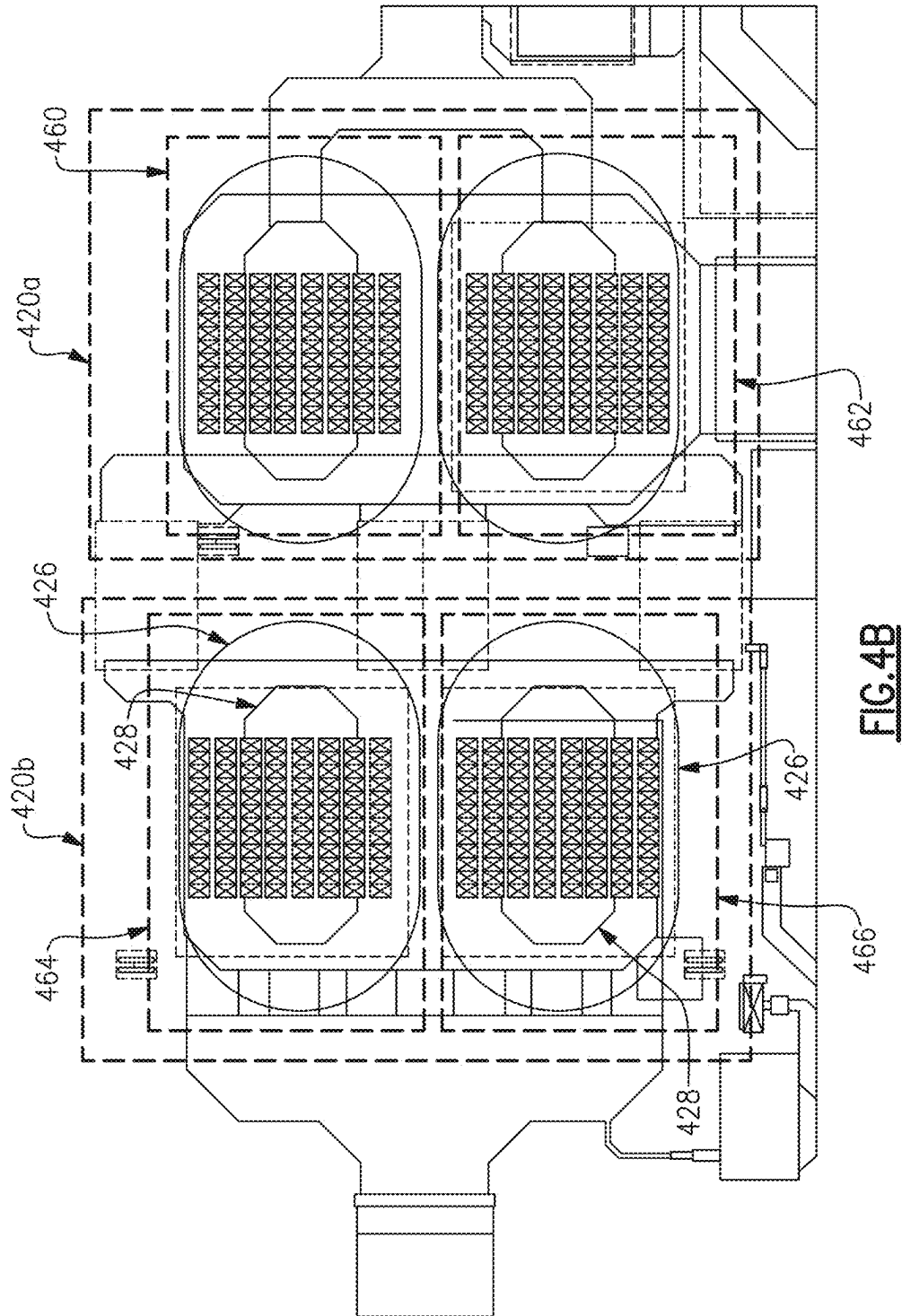
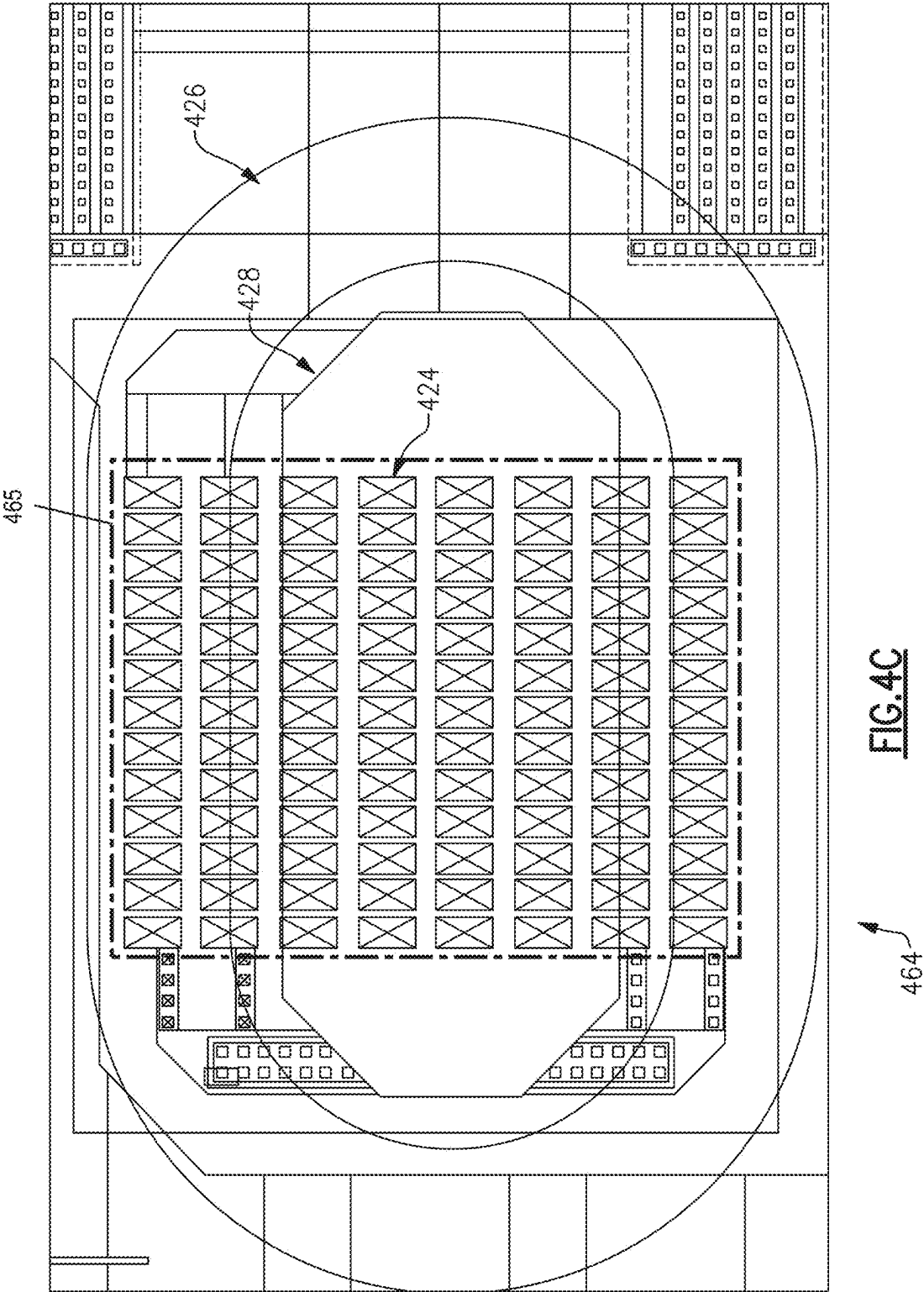


FIG. 4AA





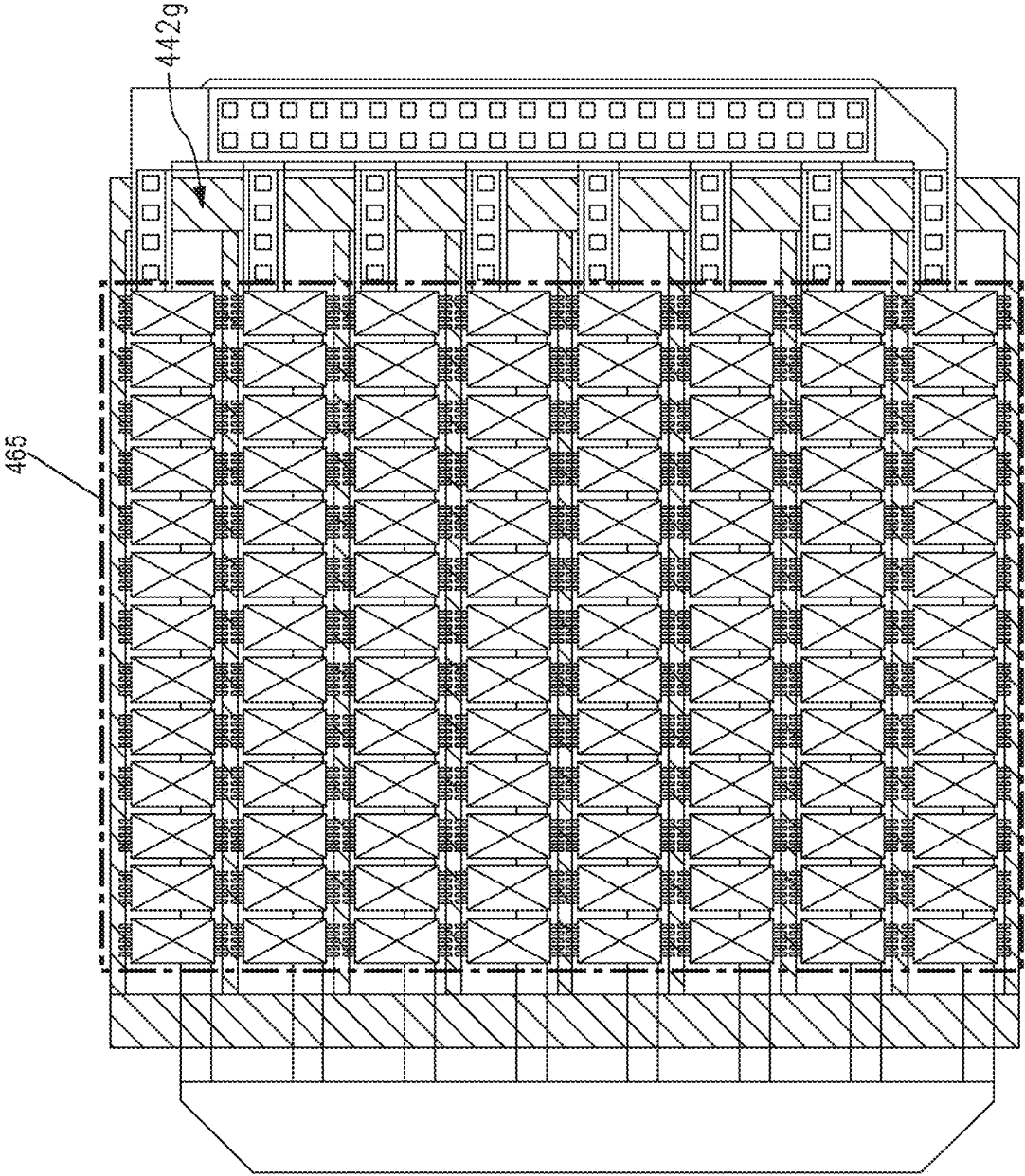


FIG. 4D

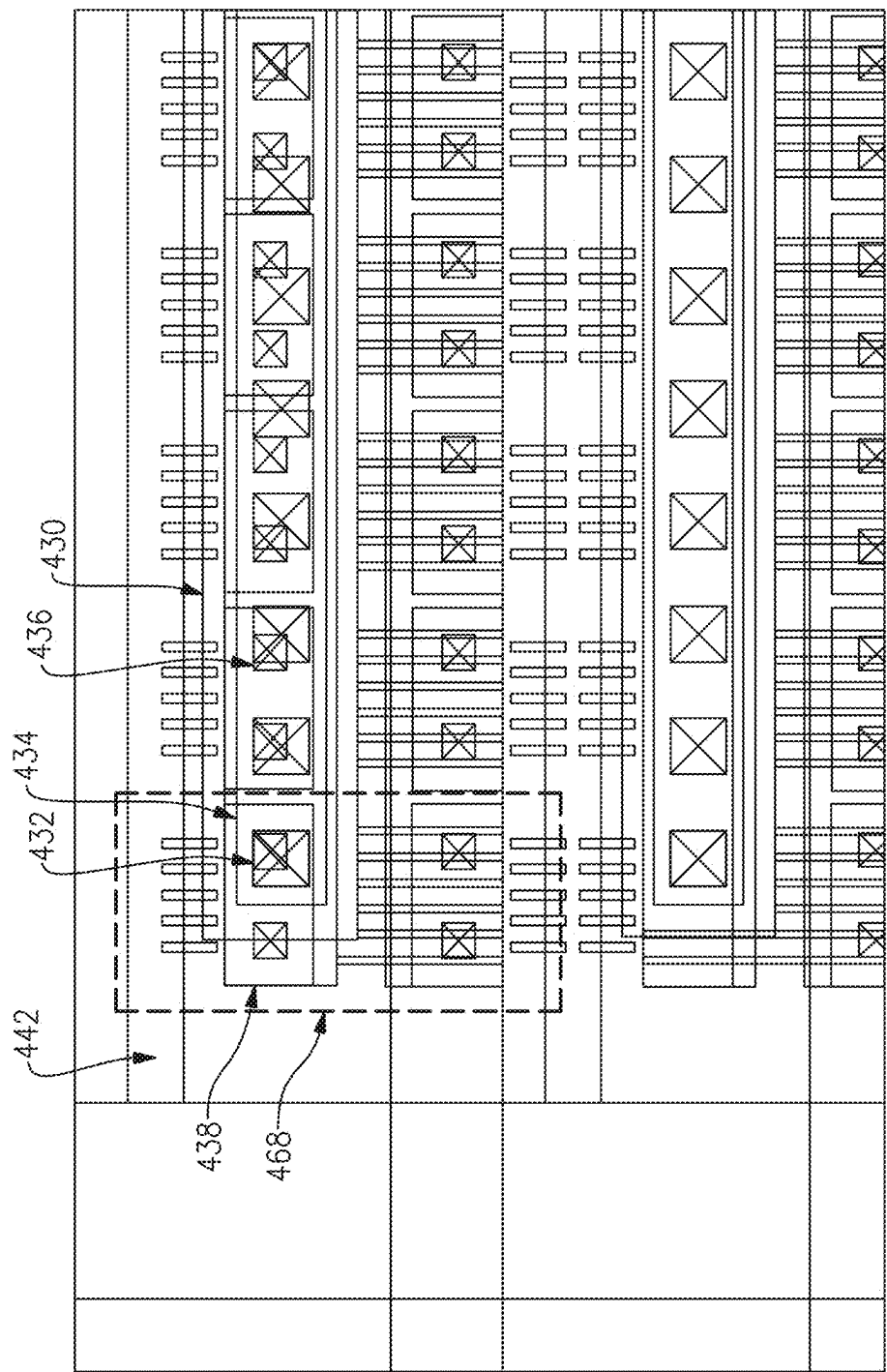


FIG. 4E

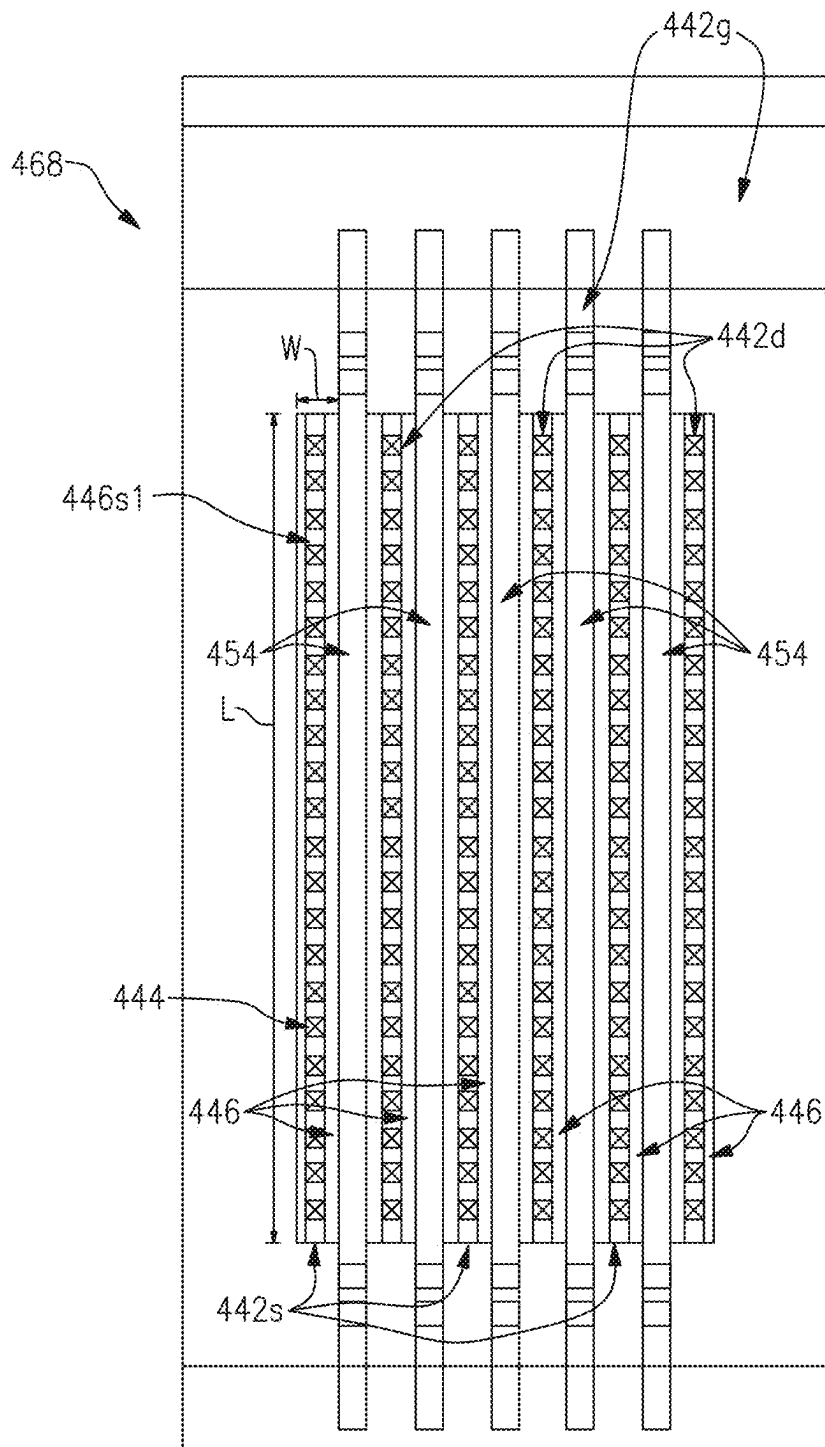


FIG. 4F

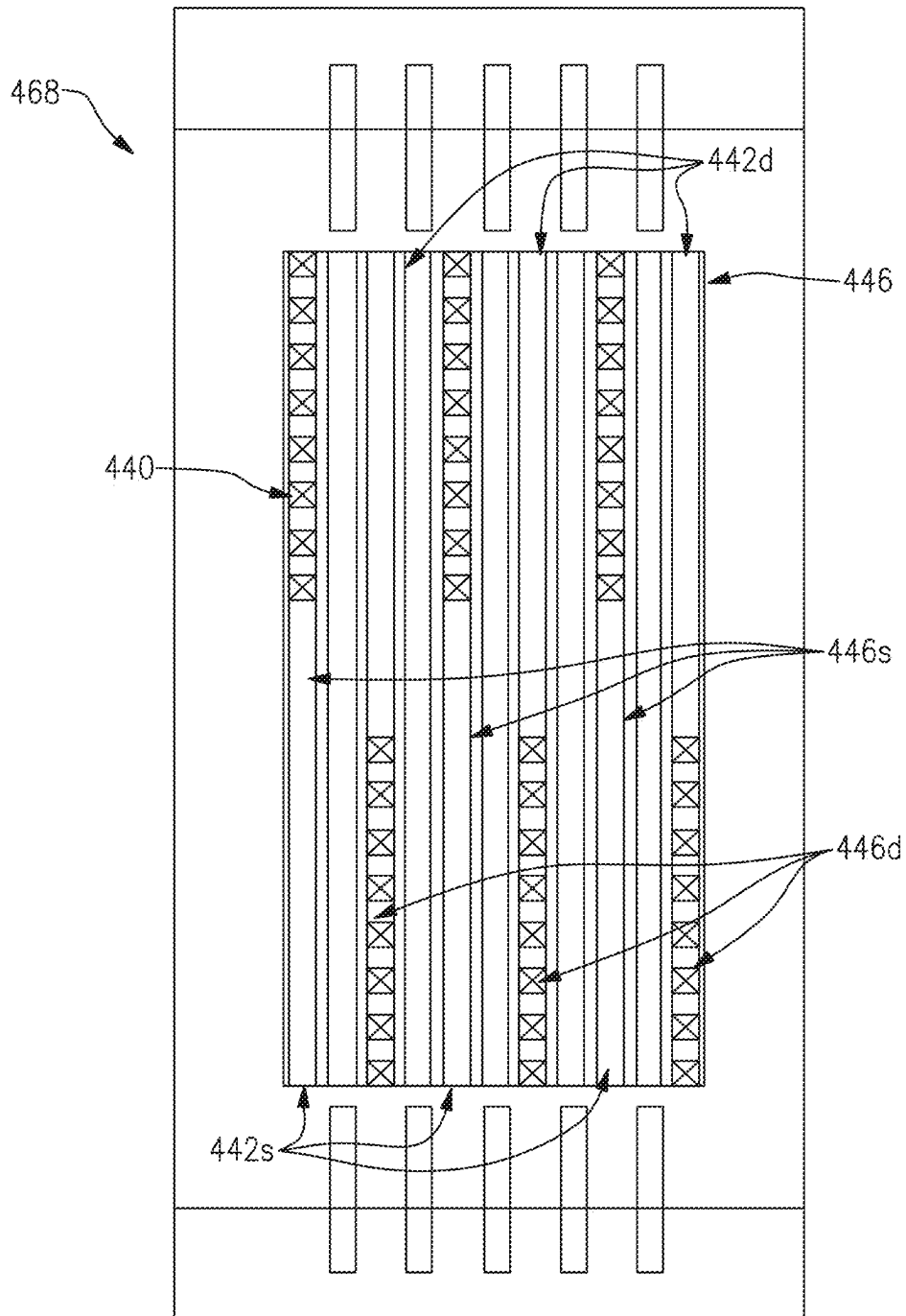


FIG. 4G

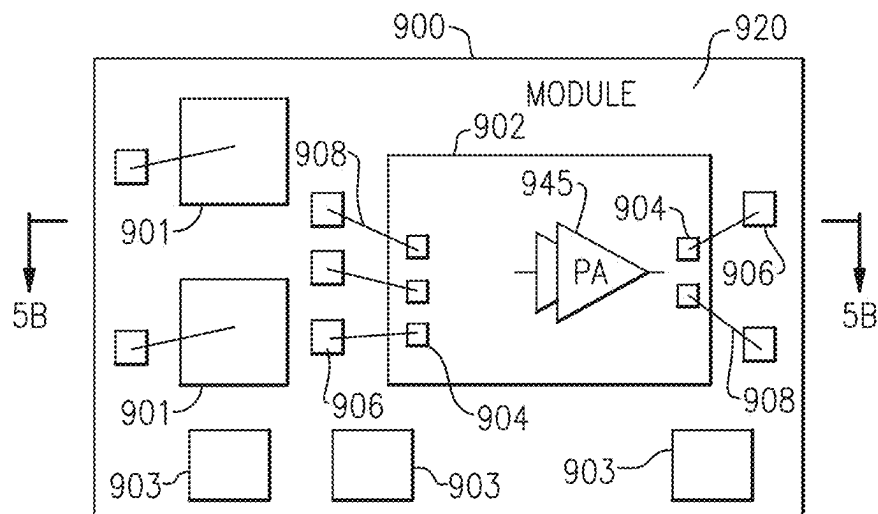


FIG. 5A

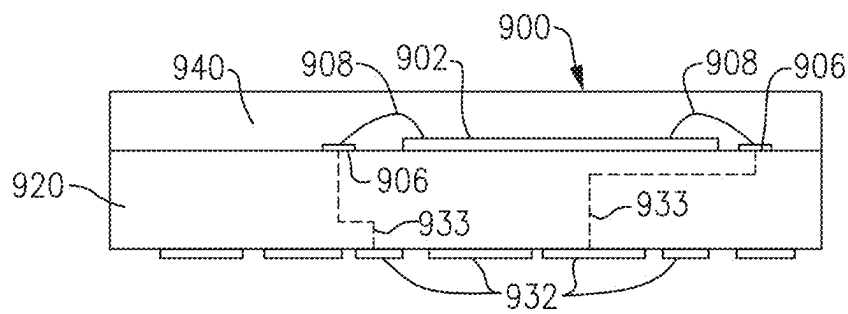


FIG. 5B

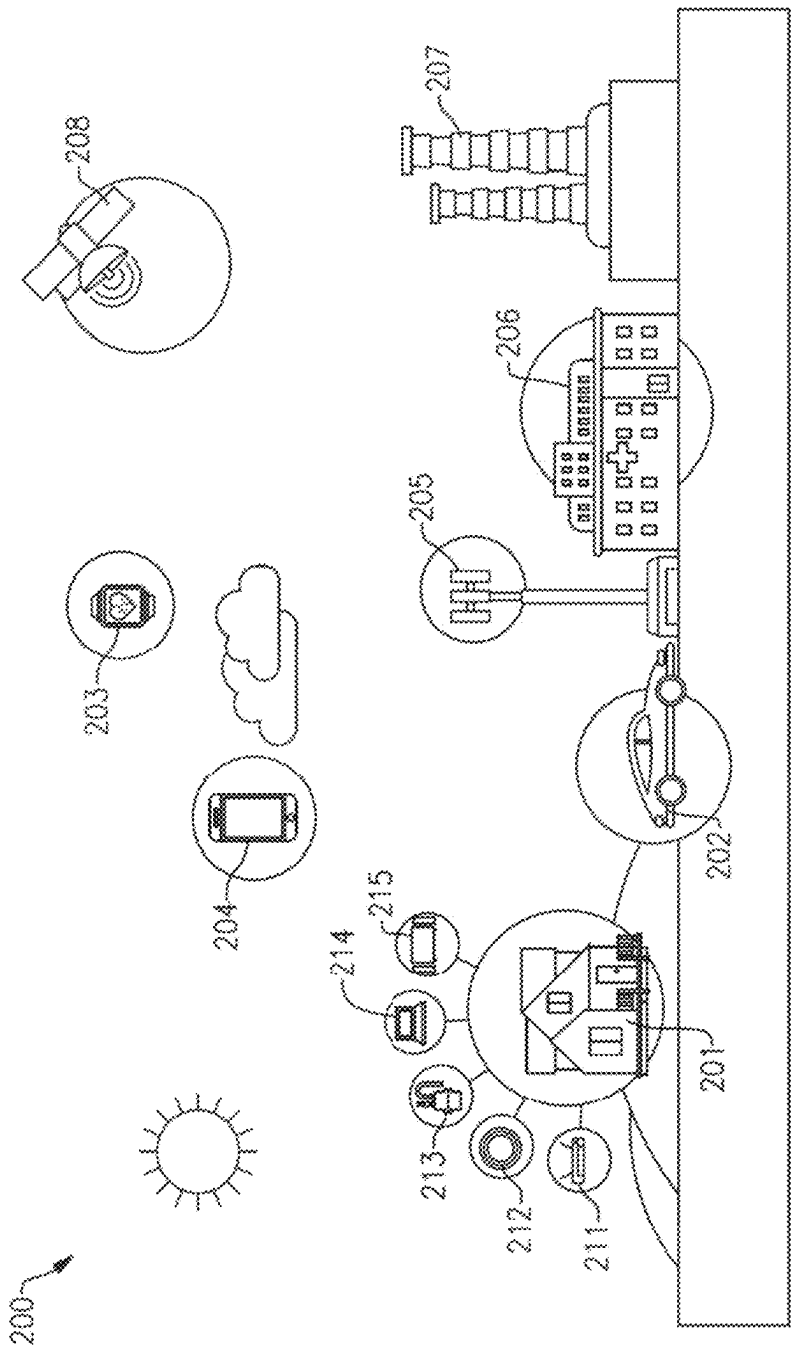


FIG. 6

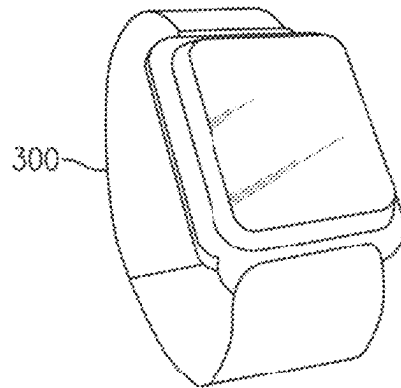


FIG. 7A

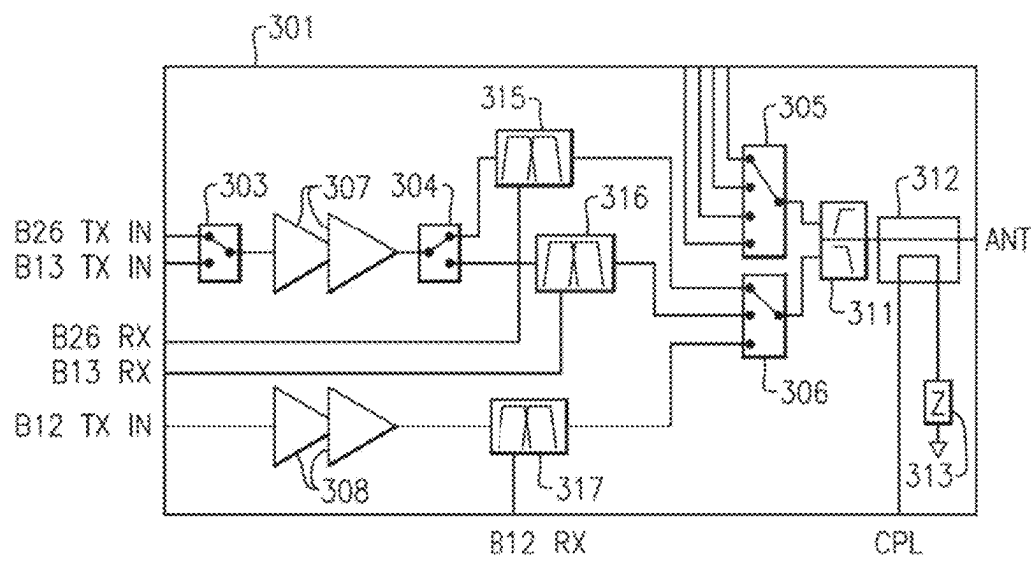


FIG. 7B

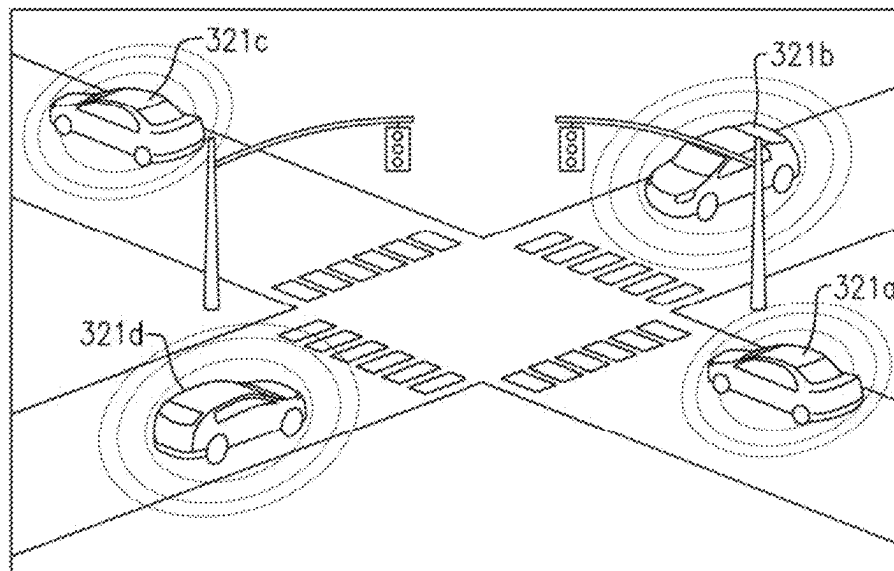


FIG. 8A

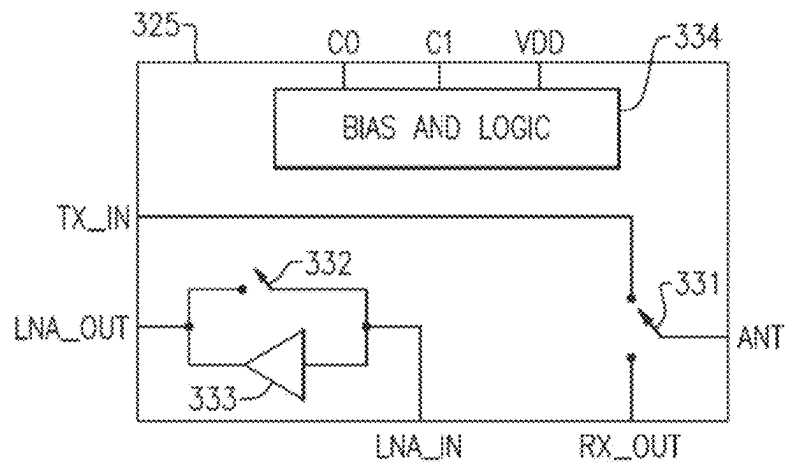


FIG. 8B

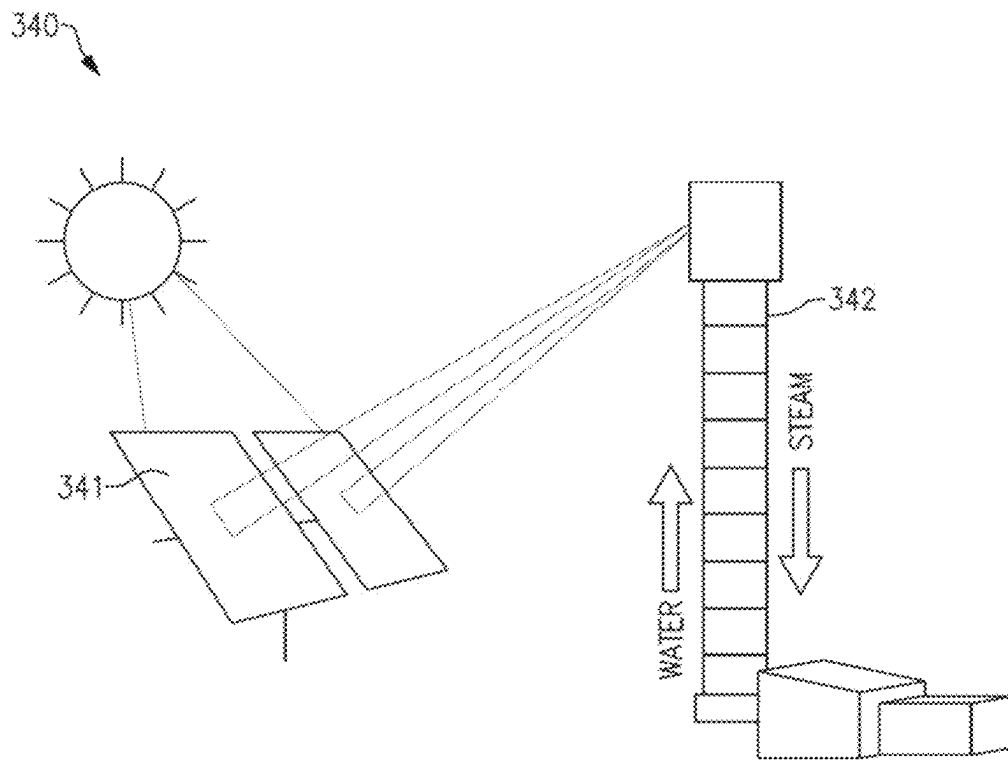


FIG. 9A

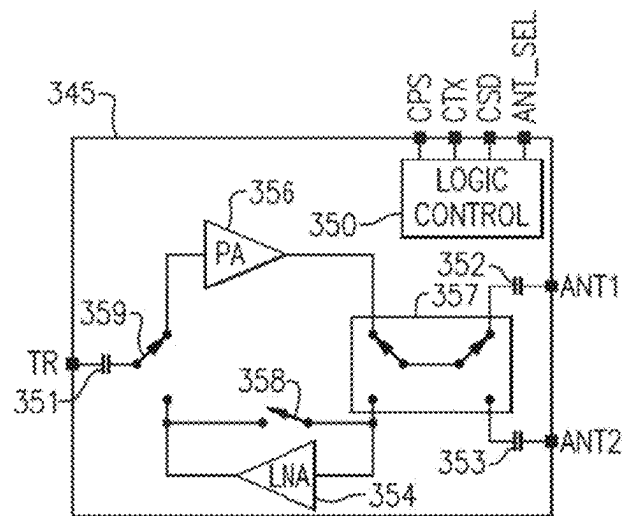


FIG. 9B

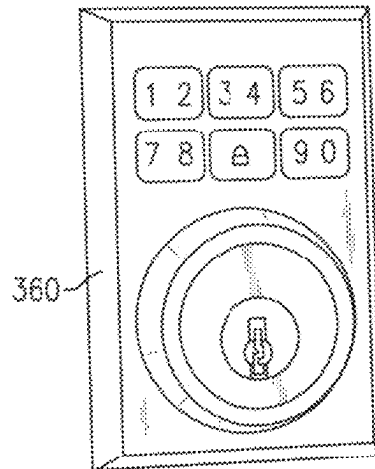


FIG. 10A

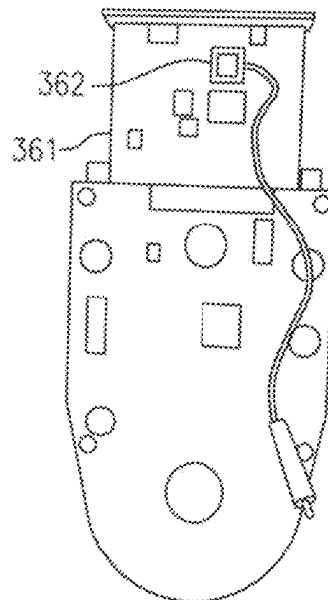


FIG. 10B

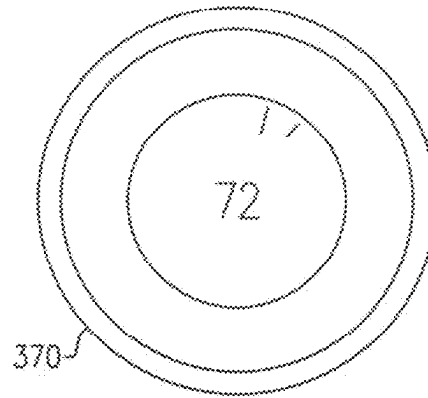


FIG. 11A

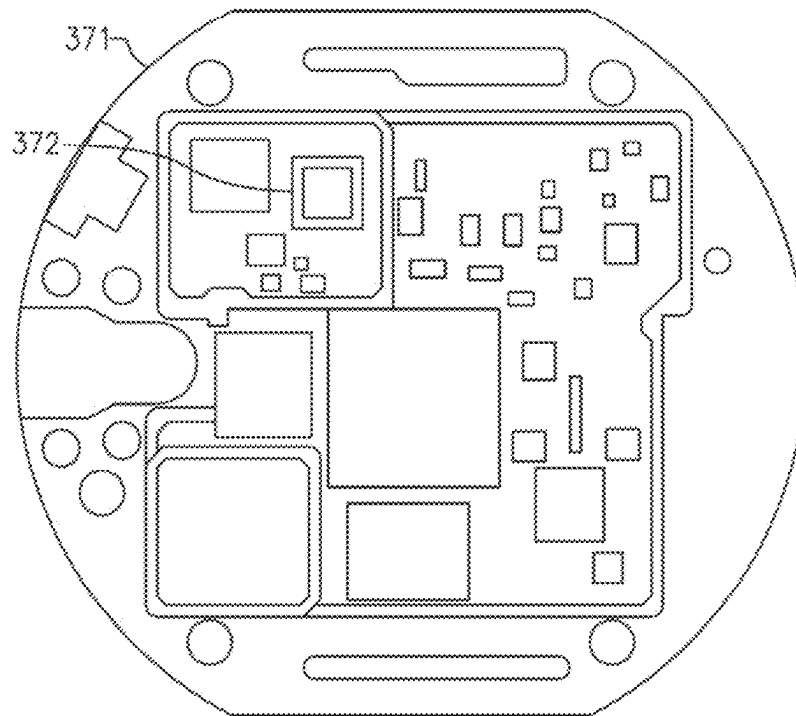


FIG. 11B

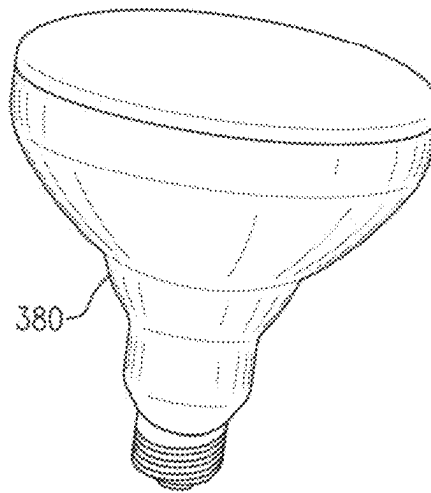


FIG. 12A

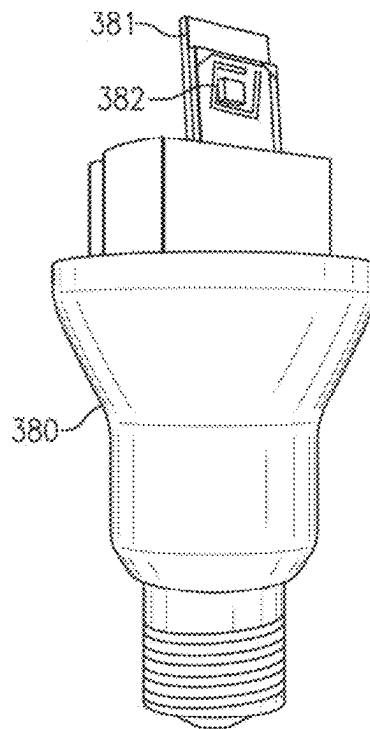


FIG. 12B

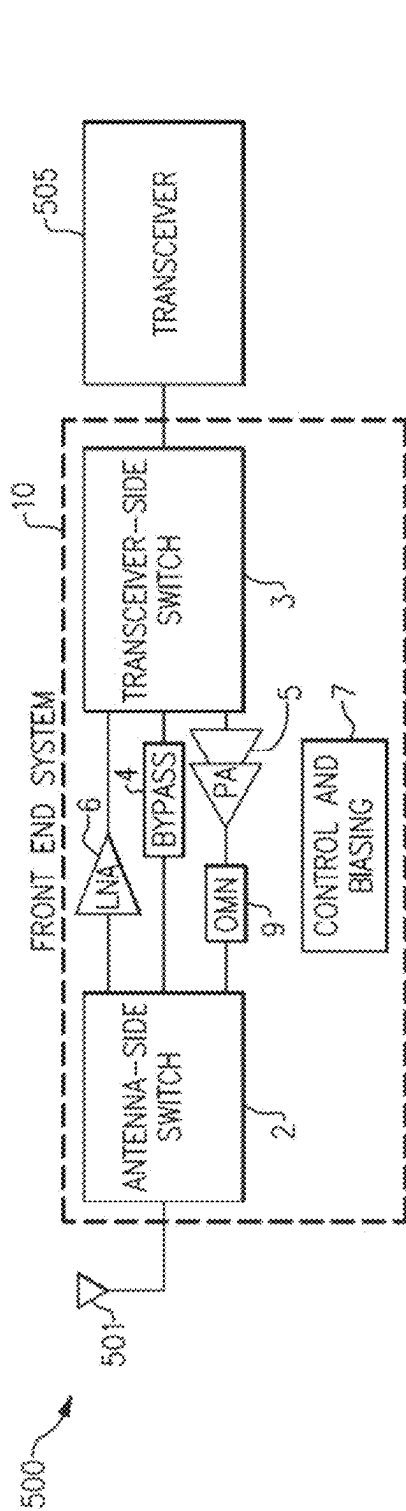


FIG. 13A

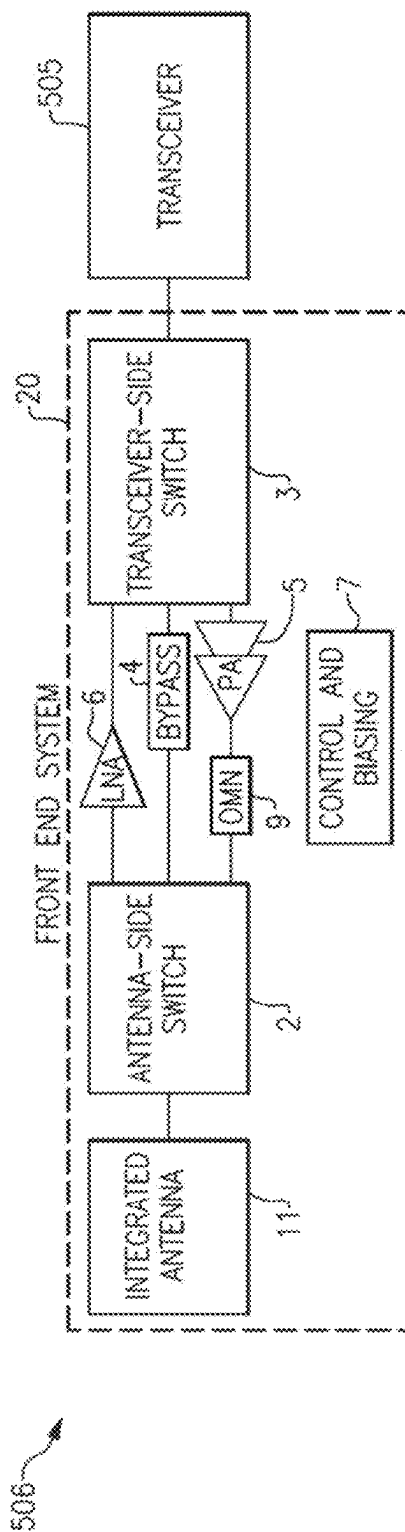


FIG. 13B

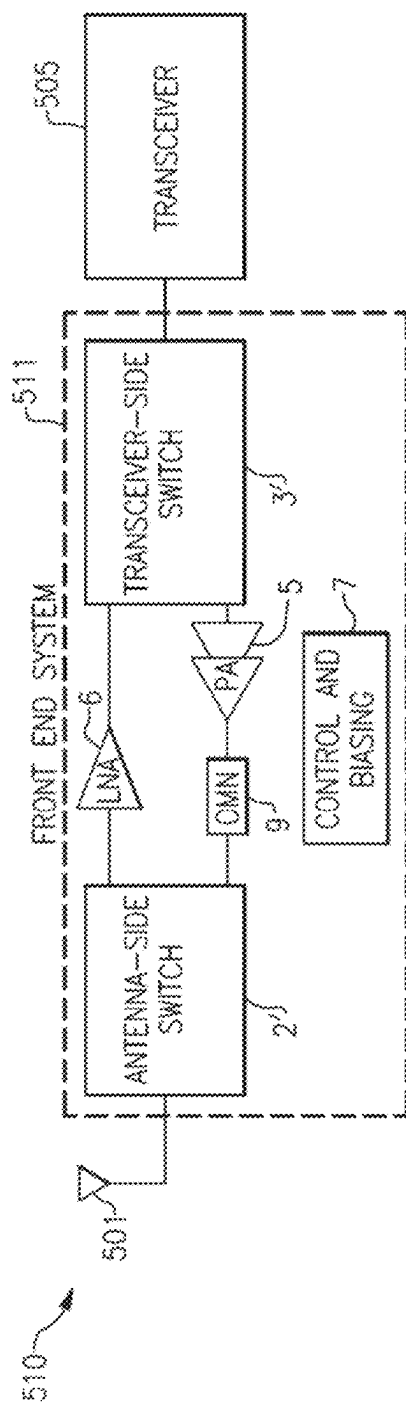


FIG. 13C

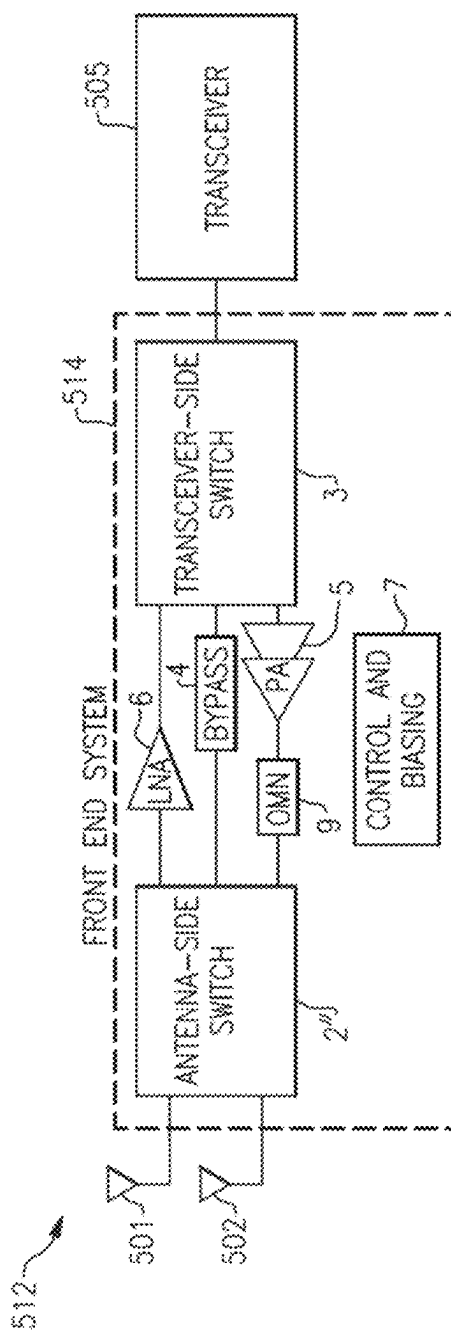
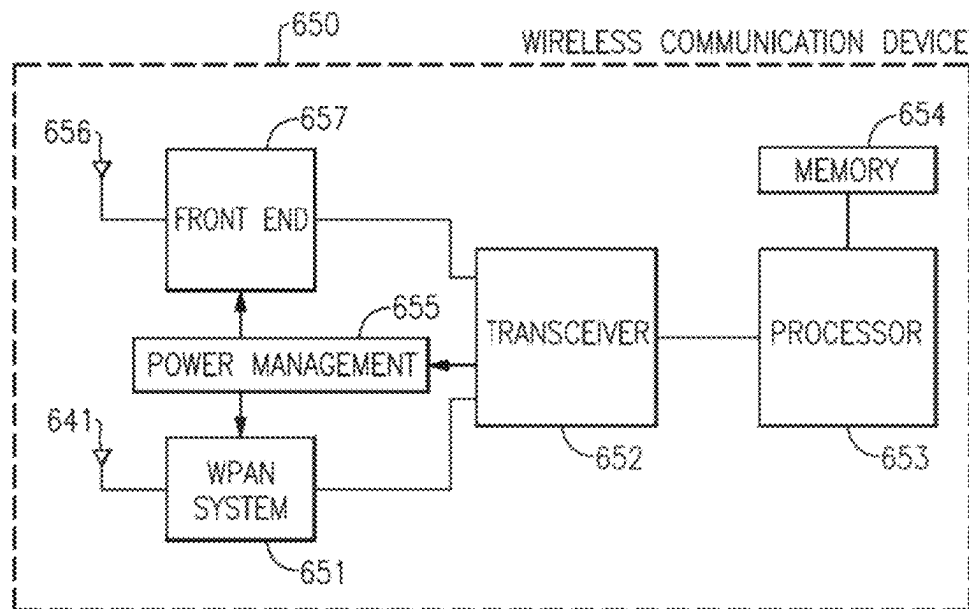
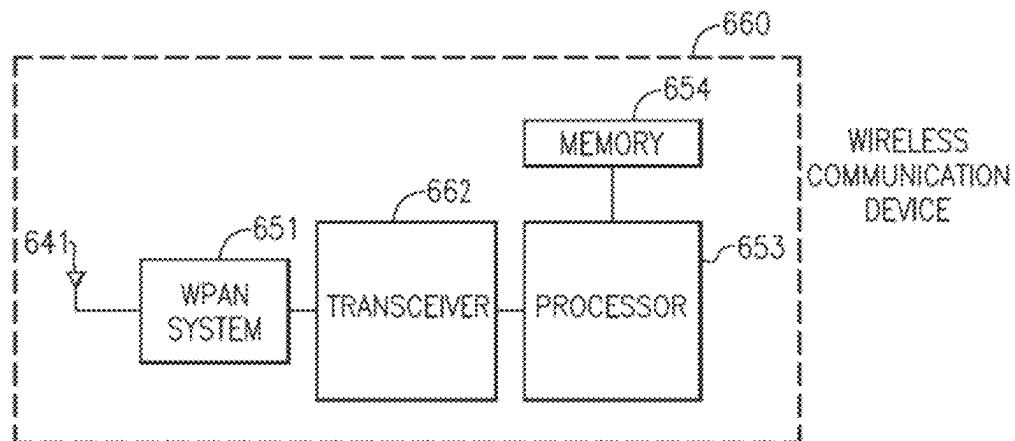


FIG. 13D

FIG.14AFIG.14B

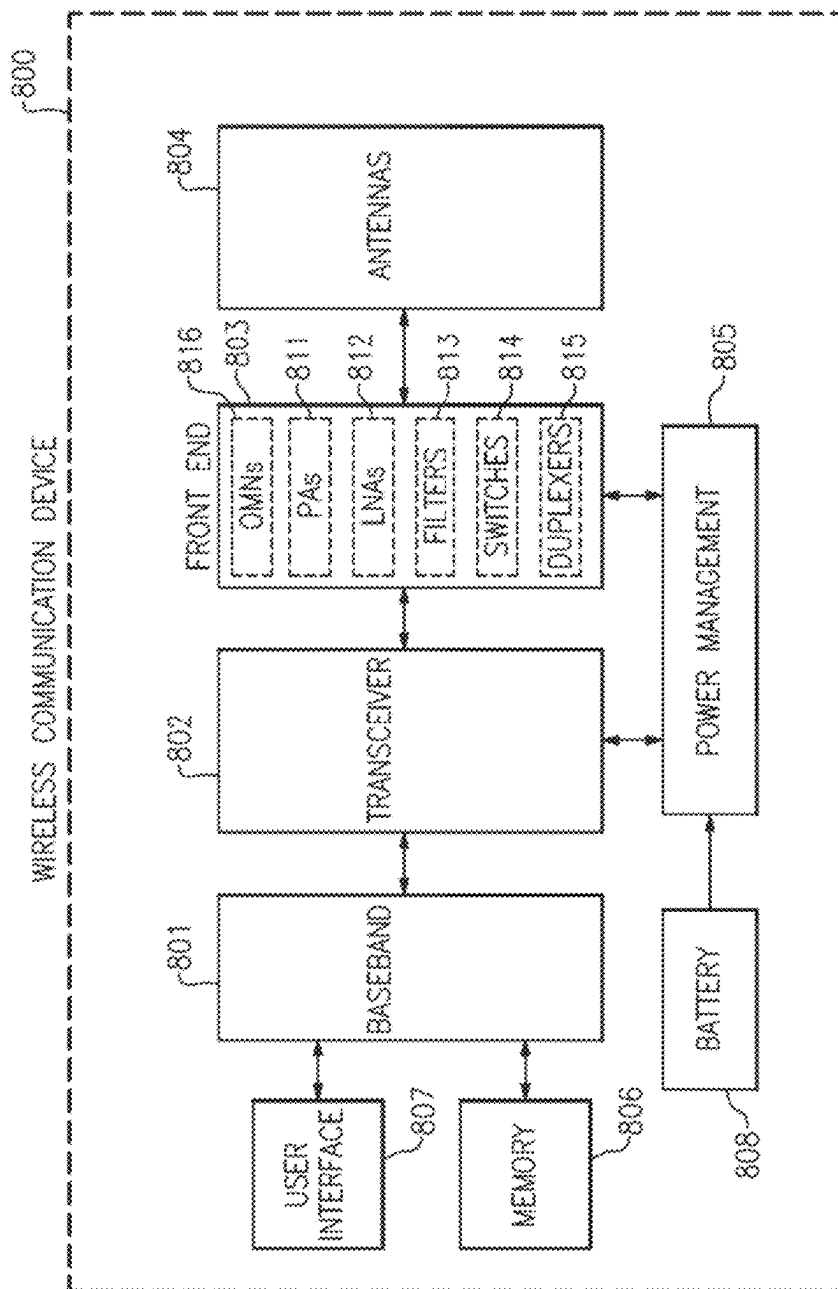


FIG.14C

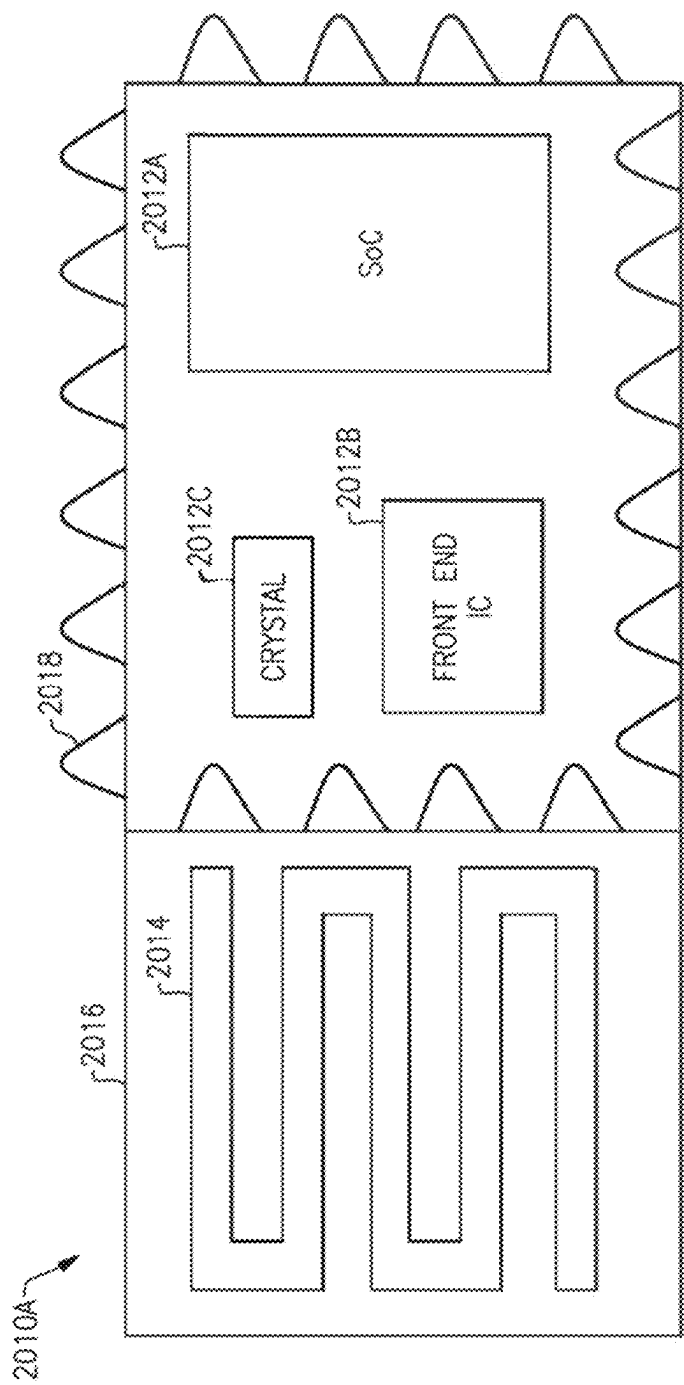


FIG.15

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FLIP-CHIP SEMICONDUCTOR-ON-INSULATOR TRANSISTOR LAYOUT

INCORPORATION BY REFERENCE TO ANY PRIORITY APPLICATIONS

Any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application are hereby incorporated by reference under 37 CFR 1.57.

BACKGROUND

Technical Field

Embodiments of this disclosure relate to radio frequency electronic systems, such as front end systems and related devices, integrated circuits, modules, and methods.

Description of Related Technology

A radio frequency electronic system can process radio frequency signals in a frequency range from about 30 kilohertz (kHz) to 300 gigahertz (GHz), such as in a range from about 450 megahertz (MHz) to 6 GHz. A front end system is an example of a radio frequency electronic system. A front end system can be referred to as a radio frequency front end system. A front end system can process signals being transmitted and/or received via one or more antennas. For example, a front end system can include one or more switches, one or more filters, one or more low noise amplifiers, one or more power amplifiers, other circuitry, or any suitable combination thereof in one or more signal paths between one or more antennas and a transceiver. Front end systems can include one or more receive paths and one or more transmit paths.

A front-end system can include a power amplifier in a transmit path. Power amplifiers can be included in front-end systems in a wide variety of communications devices to amplify an RF signal for transmission. An RF signal amplified by a power amplifier can be transmitted via an antenna. Example communications devices having power amplifiers include, but are not limited to, Internet of Things (IoT) devices, mobile phones, tablets, base stations, network access points, laptops, computers, and televisions. As an example, in mobile phones that communicate using a cellular standard, a wireless local area network (WLAN) standard, and/or any other suitable communication standard, a power amplifier can be used to amplify the RF signal.

Some such radio frequency power amplifiers are implemented in a semiconductor-on-insulator process technology, such as a silicon-on-insulator (SOI) complementary metal oxide semiconductor (CMOS) process technology.

SUMMARY

According to certain aspects of the disclosure, a semiconductor-on-insulator die or other device includes a substrate layer, an active layer, an insulator layer between the substrate layer and the active layer, a first metal layer, and a first via layer between the active layer and the first metal layer. The die can further include at least one contact pad, a transistor including a first terminal, the first terminal formed within the active layer, and a conduction path including a plurality of first conduction path portions. The first conduction path portions can extend between the first terminal and

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the at least one contact pad and reside within a footprint of the at least one contact pad. The conductive path can further include a plurality of second conduction path portions extending between the first terminal and the at least one contact pad and which reside at least partly outside of the footprint of the at least one contact pad.

The conduction path can include a first via segment formed within the first via layer and a first metal segment formed within the first metal layer. A surface area of an interface between the first via segment and the first terminal can be at least 4, 5, 6, 7, or 8 percent of the surface area of the first terminal. The first terminal substantially can completely overlap the first via segment.

The die can include a second metal layer and a second via layer. The second via layer can be between the first metal layer and the second metal layer, where the conduction path includes a second via segment formed from the second via layer and a second metal segment formed from the second metal layer. The surface area of an interface between the first metal segment and the second via segment is at least 10, 15, 20, 25, or 27 percent of the surface area of the first metal segment.

The transistor can be a field effect transistor and the first terminal can be a source of the transistor. In some embodiments, the transistor includes first and second sections, the at least one contact pad includes a first contact pad and a second contact pad, a footprint of the first contact pad overlaps with a footprint of the first section of the transistor, and a footprint of the second contact pad overlaps with a footprint of the second section of the transistor.

The die can be a flip-chip die and the first contact pad can include at least one contact bump. The footprint of the at least one contact pad can overlap with a footprint of the transistor. At least 25, 30, 40, 45, or 50 percent of the first terminal falls within the footprint of the at least one contact pad.

A packaged module can implement the die or other device, where a module metal layer can be electrically connected to the at least one contact pad of the die and a ground plane can be electrically connected to the module metal layer. The first conduction path portions can extend between the first terminal and the ground plane and residing within a footprint of the at least one contact pad. The packaged module can further include a module via layer between the module metal layer and the ground plane, and a module via segment formed within the module via layer, where the conduction path further includes the module via segment. In some embodiments, substantially the entire first terminal falls within a footprint of the module metal segment. The at least one contact pad can include first and second contact pads, and the module metal segment can include a first portion connected to the first contact pad and a second portion connected to the second contact pad, where the footprint of the module metal segment includes a footprint of the first portion and a footprint of the second portion.

A power amplifier system can implement the die or other device, where a power amplifier configured to amplify a radio frequency signal includes the first transistor. The power amplifier system can include a second transistor formed in the die. The first and/or second transistors can be field effect transistors. The first transistor can be configured in a common source configuration and the second transistor can be configured in a common gate configuration.

According to additional aspects, a flip-chip semiconductor-on-insulator die or other device can include a substrate layer, an active layer, an insulator layer between the substrate layer and the active layer, a first metal layer, and a first

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via layer between the active layer and the first metal layer. The die can further include at least first and second contact pads and a transistor including a first terminal. The first terminal can be formed within the active layer, first portion of the first terminal falling within a footprint of the first contact pad and a second portion of the first terminal falling within a footprint of the second contact pad.

At least 25, 30, 40, 45, or 50 percent of the first terminal can fall within the combined footprints of the at least first and second contact pads.

The transistor can be divided into generally equally sized portions. Each generally equally sized portion can be connected to a respective one of the at least first and second contact pads.

The die can include a conduction path extending between the first terminal and the first and second contact pads, and including a first via segment formed within the first via layer and a first metal segment formed within the first metal layer. The die can further include a second metal layer and a second via layer, the second via layer between the first metal layer and the second metal layer, and the conduction path further including a second via segment formed from the second via layer and a second metal segment formed from the second metal layer. A portion of the conduction path including the first terminal, the first via segment, the first metal segment, the second via segment, and the second metal segment extends substantially completely orthogonal to a plane that is substantially parallel to the substrate layer. The surface area of an interface between the first metal segment and the second via segment can be at least 10, 15, 20, 25, or 27 percent of the surface area of the first metal segment. The surface area of an interface between the first via segment and the first terminal is at least 4, 5, 6, 7, or 8 percent of the surface area of the first terminal.

The transistor can be a field effect transistor, where the first terminal is a source of the transistor. The at least first and second contact pads can be ground pads.

The transistor can include a first part including the first portion of the first terminal and a second part including the second portion of the first terminal. A footprint of the first part of the transistor can overlap the footprint of the first contact pad and a footprint of the second part of the transistor can overlap the footprint of the second contact pad.

The flip-chip die or other device can be incorporated into a packaged module, where a module metal layer electrically connected to the first and second contact pads of the die is electrically connected to a ground plane. The conduction path can extend between the first terminal and the ground plane, where the conduction path including a first via segment formed within the first via layer, a first metal segment formed within the first metal layer, and a module metal segment formed within the module metal layer. The first portion of the first terminal can fall within a footprint of a first portion of the module metal segment and the second portion of the first terminal can fall within a footprint of a second portion of the module metal segment. The transistor can include a first part including the first portion of the first terminal and a second part including the second portion of the first terminal. A footprint of the first part of the transistor can overlap the footprint of the first contact pad and a footprint of the second part of the transistor can overlap the footprint of the second contact pad.

The flip-chip die or other device can be incorporated into a power amplifier system, including a power amplifier configured to amplify a radio frequency signal and including the first transistor formed in the semiconductor-on-insulator

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die. The power amplifier system can include a second transistor formed in the semiconductor-on-insulator die. The second transistor can be a field effect transistor. The first transistor can be configured in a common source configuration and the second transistor can be configured in a common gate configuration.

According to certain aspects of the disclosure, a semiconductor-on-insulator die or other device can include a substrate layer, an active layer, an insulator layer between the substrate and active layers, a first metal layer, and a first via layer between the active layer and the first metal layer. The die or other device can further include at least one contact pad and a transistor including a first terminal, the first terminal formed within the active layer. A conduction path can extend between the first terminal and the at least one contact pad, where the conduction path includes a first via segment formed within the first via layer and a first metal segment formed within the first metal layer.

The transistor can be a field effect transistor, where the first terminal is a source of the transistor. The die or other device can be a flip-chip die and at least one contact pad can include a contact bump.

The surface area of an interface between the first via segment and the first terminal can be at least 4, 5, 6, 7, or 8 percent of the surface area of the first terminal. The first terminal can substantially completely overlap or completely overlap the first via segment.

A portion of the conduction path including the first terminal, the first via segment, and the first metal segment can extend substantially completely orthogonal, or completely orthogonal, to a plane that is substantially parallel, or parallel, to the substrate layer.

The die or other device can further comprise a second metal layer and a second via layer, the second via layer between the first metal layer and the second metal layer, where the conduction path further includes a second via segment formed from the second via layer and a second metal segment formed from the second metal layer. A portion of the conduction path including the first terminal, the first via segment, the first metal segment, the second via segment, and the second metal segment can extend substantially completely orthogonal, or completely orthogonal, to a plane that is substantially parallel, or parallel, to the substrate layer. The surface area of an interface between the first metal segment and the second via segment can be at least 10, 15, 20, 25, or 27 percent of the surface area of the first metal segment.

At least some portion of the conduction path can extend along the entirety of a line extending from the first terminal to the at least one contact pad, where the line is orthogonal to a plane that is substantially parallel, or parallel, to the substrate layer.

At least a portion of the first terminal can fall within a footprint of the at least one contact pad. For example, at least 25, 30, 40, 45, or 50 percent of the first terminal can fall within a footprint of the at least one contact pad. The at least one contact pad can include a first contact pad overlapping with a first portion of the transistor and a second contact pad overlapping with a second portion of the transistor, where the footprint of the at least one contact pad includes a footprint of the first contact pad and a footprint of the second contact pad.

A footprint of the first contact pad can overlap with a footprint of the transistor. The transistor can include first and second sections, where the at least one contact pad includes a first contact pad and a second contact pad. A footprint of the first contact pad can overlap with a footprint of the first

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section of the transistor. A footprint of the second contact pad can overlap with a footprint of the second section of the transistor.

A packaged module can incorporate the die or other device, where the module includes a module metal layer electrically connected to the at least one contact pad of the die or other device. The module can include a ground plane electrically connected to the module metal layer. The conduction path can extend between the first terminal and the ground plane, where the conduction path can include the first via segment, the first metal segment, and a module metal segment formed within the module metal layer. The module can further include a module via layer between the module metal layer and the ground plane, and a module via segment formed within the module via layer, where the conduction path can further include the module via segment.

Substantially the entire first terminal can fall within a footprint of the module metal segment. The at least one contact pad can include first and second contact pads, where the module metal segment includes a first portion connected to the first contact pad and a second portion connected to the second contact pad. The footprint of the module metal segment can include a footprint of the first portion and a footprint of the second portion.

A power amplifier system can include the semiconductor-on-insulator die or other device, where the power amplifier system includes a power amplifier configured to amplify a radio frequency signal. The power amplifier can include the transistor. The power amplifier can further include a second transistor formed in the semiconductor-on-insulator die or other device, which can also be a field effect transistor. The transistor can be configured in a common source configuration and the second transistor can be configured in a common gate configuration.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1A illustrates a schematic block diagram of one example of a front end system.

FIG. 1B illustrates a schematic block diagram of another example of a front end system.

FIG. 2 is a schematic diagram showing portions of a transmit path of a front end system including an example of an output matching network.

FIG. 3 is a schematic diagram showing portions of a transmit path of a front end system including another example of an output matching network.

FIG. 4 depicts an example of a mobile device including a radio frequency module having a semiconductor-on-insulator die.

FIGS. 4A and 4AA are cross-sectional and perspective schematic views of a portion of a semiconductor-on-insulator die connected to a module via a flip-chip connection.

FIG. 4B is a top view showing certain layers of a portion of a semiconductor-on-insulator die, depicting transistors of a cascode power amplifier implemented on the die.

FIGS. 4C-4G are zoomed in views showing certain layers of the die of FIG. 4B.

FIG. 5A is a schematic diagram of one embodiment of a packaged module.

FIG. 5B is a schematic diagram of a cross-section of the packaged module of FIG. 5A taken along the lines 5B-5B.

FIG. 6 is a schematic diagram of one example of an Internet of things (IoT) network.

FIG. 7A is a schematic diagram of one example of an IoT-enabled watch.

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FIG. 7B is a schematic diagram of one example of a front end system for an IoT-enabled object.

FIG. 8A is a schematic diagram of one example of IoT-enabled vehicles.

FIG. 8B is a schematic diagram of another example of a front end system for an IoT-enabled object.

FIG. 9A is a schematic diagram of one example of IoT-enabled industrial equipment.

FIG. 9B is a schematic diagram of another example of a front end system for an IoT-enabled object.

FIG. 10A is a schematic diagram of one example of an IoT-enabled lock.

FIG. 10B is a schematic diagram of one example of a circuit board for the IoT-enabled lock of FIG. 10A.

FIG. 11A is a schematic diagram of one example of an IoT-enabled thermostat.

FIG. 11B is a schematic diagram of one example of a circuit board for the IoT-enabled thermostat of FIG. 11A.

FIG. 12A is a schematic diagram of one example of IoT-enabled light.

FIG. 12B is a schematic diagram of one example of a circuit board for the IoT-enabled light of FIG. 12A.

FIG. 13A illustrates a schematic block diagram of one example of a radio frequency system.

FIG. 13B illustrates a schematic block diagram of another example of a radio frequency system.

FIG. 13C illustrates a schematic block diagram of another example of a radio frequency system.

FIG. 13D illustrates a schematic block diagram of another example of a radio frequency system.

FIG. 14A is a schematic diagram of one example of a wireless communication device.

FIG. 14B is a schematic diagram of another example of a wireless communication device.

FIG. 14C is a schematic diagram of another example of a wireless communication device.

FIG. 15 is a schematic diagram of an example of a radio frequency module according to an embodiment.

DETAILED DESCRIPTION OF CERTAIN EMBODIMENTS

The following detailed description of certain embodiments presents various descriptions of specific embodiments. However, the innovations described herein can be embodied in a multitude of different ways, for example, as defined and covered by the claims. In this description, reference is made to the drawings where like reference numerals can indicate identical or functionally similar elements. It will be understood that elements illustrated in the figures are not necessarily drawn to scale. Moreover, it will be understood that certain embodiments can include more elements than illustrated in a drawing and/or a subset of the elements illustrated in a drawing. Further, some embodiments can incorporate any suitable combination of features from two or more drawings.

Front End Systems

A front end system can be used to handle signals being transmitted and/or received via one or more antennas. For example, a front end system can include switches, filters, amplifiers, and/or other circuitry in signal paths between one or more antennas and a transceiver.

Implementing one or more features described herein in a front end system can achieve a number of advantages, including, but not limited to, one or more of higher power added efficiency (PAE), more compact layout, lower cost, higher linearity, superior robustness to overstress, and/or

enhanced integration. Moreover, implementing one or more features described herein in a front end system can achieve desirable figure of merit (FOM) and/or other metrics by which front end systems are rated. Although some features are described herein in connection with front end systems for illustrative purposes, it will be understood that the principles and advantages described herein can be applied to a wide variety of other electronics.

FIG. 1A illustrates a schematic block diagram of one example of a front end system 10. The front end system 10 includes an antenna-side switch 2, a transceiver-side switch 3, a bypass circuit 4, a power amplifier 5, an output matching network 9 connected to an output of the power amplifier 5, a low noise amplifier (LNA) 6, and a control and biasing circuit 7. The front end system 10 can incorporate one or more features described in the sections herein.

Although one example of a front end system is shown in FIG. 1A, a front end system can be adapted in a wide variety of ways. For example, a front end system can include more or fewer components and/or signals paths. Accordingly, the teachings herein are applicable to front end systems implemented in a wide variety of ways.

In certain implementations, a front end system, such as the front end system 10 of FIG. 1A, is implemented on an integrated circuit or semiconductor die. In such implementations, the front end system can be referred to as a front end integrated circuit (FEIC). In other implementations, a front end system is implemented as a module. In such implementations, the front end system can be referred to as a front end module (FEM).

Accordingly, in some implementations, the front end system 10 is implemented in a packaged module. Such packaged modules can include a relatively low cost laminate and one or more dies that combine low noise amplifiers with power amplifiers and/or switch functions. Some such packaged modules can be multi-chip modules. In certain implementations, some or the all of the illustrated components of the front end system 10 can be embodied on a single integrated circuit or die. Such a die can be manufactured using any suitable process technology. As one example, the die can be a semiconductor-on-insulator die, such as a silicon-on-insulator (SOI) die, and the power amplifier 5 can include stacked field effect complementary metal oxide (CMOS) transistors. Using silicon-on-insulator or other semiconductor-on-insulator technology and stacked transistor topologies can enable power amplifiers to be implemented in relatively inexpensive and relatively reliable technology. Moreover, the desirable performance of low-noise amplifiers (LNAs) and/or multi-throw RF switches in silicon-on-insulator technology can enable a stacked transistor silicon-on-insulator power amplifier to be implemented as part of a complete front end integrated circuit (FEIC) solution that includes transmit, receive, and switching functionality with desirable performance.

For example, in some embodiments the front end system 10 can be a packaged module including a semiconductor-on-insulator die implementing a FET-based class-F power amplifier 5 and an output matching network 9. Further details regarding such embodiments, and exemplary output matching networks are described in further detail herein, such as with respect to FIGS. 2 and 3, for example.

As shown in FIG. 1A, the front end system 10 includes multiple signal paths between the antenna-side switch 2 and the transceiver-side switch 3. For example, the illustrated front end system 10 includes a bypass signal path that includes the bypass circuit 4, a transmit signal path that includes the power amplifier 5, and a receive signal path that

includes the LNA 6. Although an example with three signal paths is shown, a front end system can include more or fewer signal paths.

The antenna-side switch 2 is used to control connection of the signal paths to an antenna (not shown in FIG. 1A). For example, the antenna-side switch 2 can be used to connect a particular one of the transmit signal path, the receive signal path, or the bypass signal path to an antenna. Additionally, the transceiver-side switch 3 is used to control connection of the signal paths to a transceiver (not shown in FIG. 1A). For example, the transceiver-side switch 3 can be used to connect a particular one of the transmit signal path, the receive signal path, or the bypass signal path to a transceiver. In certain implementations, the antenna-side switch 2 and/or the transceiver-side switch 3 are implemented as multi-throw switches.

FIG. 1B illustrates a schematic block diagram of another example of a front end system 20. The front end system 20 of FIG. 1B is similar to the front end system 10 of FIG. 1A, except that the front end system 20 further includes an integrated antenna 11. In certain implementations, a front end system includes an integrated antenna. For example, a front end system can be implemented on a module along with one or more integrated antennas.

With reference to FIGS. 1A and 1B, the bypass network 4 can include any suitable network for matching and/or bypassing the receive signal path and the transmit signal path. The bypass network 4 can be implemented, for instance, by a passive impedance network or by a conductive trace or wire.

The power amplifier 5 can be used to amplify a transmit signal received from a transceiver for transmission via an antenna. The power amplifier 5 can be implemented in a wide variety of ways.

In certain implementations, the power amplifier 5 is a class F amplifier including two field-effect transistors implemented in a cascode arrangement (see, e.g., FIG. 3). Such a stacked power amplifier topology can be advantageous in semiconductor-on-insulator process technologies. For instance, device stacking for silicon-on-insulator power amplifier circuit topologies can overcome relatively low breakdown voltages of scaled transistors. Such device stacking can be beneficial in applications in which a stacked amplifier is exposed to a relatively large voltage swing, such as a voltage swing exceeding about 2.75 Volts. Stacking several transistors, such as 2, 3, 4 or more transistors, can result in a power amplifier with desirable operating characteristics.

In additional embodiments, the power amplifier 5 can include a stacked output stage and a bias circuit that biases the stacked transistors of the stacked output stage based on mode. In one example, the bias circuit can bias a transistor in a stack to a linear region of operation in a first mode, and bias the transistor as a switch in a second mode. Accordingly, the bias circuit can bias the stacked output stage such that the stacked output stage behaves like there are fewer transistors in the stack in the second mode relative to the first mode. Such operation can result in meeting design specifications for different power modes, in which a supply voltage provided to the stacked output stage changes based on mode.

In certain implementations, the power amplifier 5 can include a driver stage implemented using an injection-locked oscillator and an output stage having an adjustable supply voltage that changes with a mode of the power amplifier 5. By implementing the power amplifier 5 in this manner, the power amplifier 5 exhibits excellent efficiency, including in a low power mode. For example, in the low

power mode, the adjustable supply voltage used to power the output stage is decreased, and the driver stage has a relatively large impact on overall efficiency of the power amplifier 5. By implementing the power amplifier 5 in this manner, the power amplifier's efficiency can be enhanced, particularly in applications in which the power amplifier's output stage operates with large differences in supply voltage in different modes of operation.

The LNA 6 can be used to amplify a received signal from the antenna. The LNA 6 can be implemented in a wide variety of ways.

In some embodiments, the LNA 6 is implemented with magnetic coupling between a degeneration inductor (e.g., a source degeneration inductor or an emitter degeneration inductor) and a series input inductor. These magnetically coupled inductors can in effect provide a transformer, with a primary winding in series with the input and a secondary winding electrically connected where the degeneration inductor is electrically connected to the amplifying device (e.g., at the source of a field effect transistor amplifying device or at the emitter of a bipolar transistor amplifying device). Providing magnetically coupled inductors in this manner allows the input match inductor to have a relatively low inductance value and corresponding small size. Moreover, negative feedback provided by the magnetically coupled inductors can provide increased linearity to the LNA 6.

In certain embodiments, the LNA 6 and the antenna-side switch 2 are implemented in accordance with one or more features of Section II (Overload Protection of Low Noise Amplifier). For example, the antenna-side switch 2 can include an analog control input for controlling an impedance between an antenna and an input to the LNA 6. Additionally, an overload protection circuit is included to provide feedback to the switch's analog control input based on detecting a signal level of the LNA 6. Thus, the overload protection circuit detects whether or not the LNA 6 is overloaded. Additionally, when the overload protection circuit detects an overload condition, the overload protection circuit provides feedback to the analog control input of the switch to increase the impedance of the switch and reduce the magnitude of the input signal received by the LNA 6. Implementing the LNA 6 and the antenna-side switch 2 in this manner limits large current and/or voltage swing conditions manifesting within amplification transistors of the LNA 6.

With continuing reference to FIGS. 1A and 1B, the control and biasing circuit 7 can be used to control and bias various front end circuitry. For example, the control and biasing circuit 7 can receive control signal(s) for controlling the LNA 6, the antenna-side switch 2, the transceiver-side switch 3, and/or the power amplifier 5. The control signals can be provided to the control and biasing circuit 7 in a variety of ways, such as over an input pad of a die. In one example, the control signals include at least one of a mode signal or a bias control signal.

The front end system 10 of FIG. 1A and the front end system 20 of FIG. 1B can be implemented on one or more semiconductor dies. In certain implementations, at least one of the semiconductor dies includes pins or pads protected using an electrical overstress (EOS) protection circuit. For example, an EOS protection circuit can include an overstress sensing circuit electrically connected between a pad of a semiconductor die and a first supply node, an impedance element electrically connected between the pad and a signal node, a controllable clamp electrically connected between the signal node and the first supply node and selectively activatable by the overstress sensing circuit, and an over-

shoot limiting circuit electrically connected between the signal node and a second supply node. The overstress sensing circuit activates the controllable clamp when an EOS event is detected at the pad. Thus, the EOS protection circuit is arranged to divert charge associated with the EOS event away from the signal node to provide EOS protection. By implementing a front end system in this manner can achieve enhanced EOS protection, lower static power dissipation, and/or a more compact chip layout. In certain implementations, the pad is an input pad that receives a control signal for controlling the power amplifier 5 and/or LNA 6.

In accordance with certain embodiments, the front end systems of FIGS. 1A and/or 1B can include RF shielding and/or RF isolation structures. For example, the front end system can be implemented as a radio frequency module that is partially shielded. Additionally, a shielding layer is included over a shielded portion of the radio frequency module and an unshielded portion of the radio frequency module is unshielded. The shielding layer can shield certain components of the front end system (for instance, the power amplifier 5 and/or LNA 6) and leave other components (for instance, the integrated antenna 11) unshielded.

In certain implementations, the front end systems of FIGS. 1A and/or 1B can include a laminated substrate including an antenna is printed on a top layer and a ground plane for shielding on a layer underneath the top layer. Additionally, at least one electronic component of the front end can be disposed along a bottom layer of the laminate substrate, and solder bumps are disposed around the electronic component and electrically connected to the ground plane. The solder bumps can attach the module to a carrier or directly to a system board. The electronic component can be surrounded by solder bumps, and the outside edges of the electronic component can have ground solder bumps that are connected to the ground plane by way of vias. Accordingly, a shielding structure with can be completed when the module is placed onto a carrier or system board, and the shielding structure can serve as a Faraday cage around the electronic component.

In certain embodiments, the front end systems disclosed herein are implemented on a semiconductor die as front end integrated circuit (FEIC). The FEIC can be included in a packaged module that stacks multiple chips and passive components, such as capacitors and resistors, into a compact area on a package substrate. By implementing a FEIC in such a packaged module, a smaller footprint and/or a more compact substrate area can be achieved.

In accordance with certain embodiments, a packaged module includes a FEIC, a crystal oscillator and a system on a chip (SoC), such as a transceiver die. The SoC can be stacked over a crystal assembly to save space and provide shorter crystal traces. The crystal assembly includes the crystal oscillator housed in a housing that includes one or more conductive pillars for routing signals from the SoC to a substrate and/or to provide thermal conductivity.

In accordance with certain embodiments, a packaged module includes a FEIC, a filter assembly and a SoC. For example, the filter assembly can be stacked with other dies and components of the packaged module to reduce a footprint of the packaged module. Furthermore, stacking the filter assembly in this manner can reduce lengths of signal carrying conductors, thereby reducing parasitics and enhancing signaling performance.

A system in a package (SiP) can include integrated circuits and/or discrete components within a common package. Some or all of a front end system can be implemented

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in a SiP. An example SiP can include a system-on-a-chip (SoC), a crystal for clocking purposes, and a front-end module (FEM) that includes a front end system. In certain SiPs, a SoC and a crystal can consume a relatively large amount of physical area. This can create a relatively large footprint for the SiP.

Power Amplifier Output Matching

FIG. 2 is a schematic diagram showing portions of transmit path of a power amplifier front end 10, including a power amplifier 5, a supply voltage biasing circuit 15, an output matching circuit 9 (also referred to as an output matching network), and at least one RF front end component 12. For example, the front end component 12 can include a switch such as the antenna-side switch 2 of FIG. 1A/1B, which connects the transmit path to an antenna 14, which can either be integrated together with the front end 10 (e.g., FIG. 1B) or external to the front end 10 (e.g., FIG. 1A).

The power amplifier 5 includes an input configured to receive a radio frequency signal RF_IN and an output electrically connected to an input node of the output matching circuit 9. The output matching circuit 9 further includes an output node electrically connected to an input of the RF front end component 12. The RF front end component 12 include an output electrically connected to the antenna 14.

The illustrated power amplifier 5 can include one or more field effect transistors implemented in a semiconductor-on-insulator die, one example of which is depicted in FIG. 3. Referring to FIGS. 2 and 3, the power amplifier illustrated in FIG. 3 includes first and second field effect transistors (FETs) 17, 19 connected in a stacked cascade configuration. For example, the first and second FETs 17, 19 can be semiconductor-on-insulator (e.g., silicon-on-insulator) transistors. Each FET includes a gate, drain, and source. The first FET 17 is connected in a common-gate configuration with the gate electrically connected to a power low supply voltage V_L , which can be a ground supply. The source of the first FET 17 is connected to the drain of the second FET 19. The second FET 19 is connected in a common-source configuration, with the source connected to the first power low supply voltage V_L . The gate of the second FET 19 is configured to receive the signal RF_IN. During operation, the power amplifier 5 can amplify the signal RF_IN, and provide the amplified signal at the drain of the first FET 17, which operates as the output of the power amplifier 5. The first and second FETs 17, 19 can be any suitable FET devices. Although FIG. 3 illustrates one implementation of the power amplifier 5, the teachings described herein can be applied to a variety of power amplifier structures, including, for example, multi-stage power amplifier structures and/or power amplifiers employing other transistor types, including, for example, bipolar transistors.

The power amplifier 5 in certain embodiments is a class F power amplifier. This can be advantageous in implementations where the power amplifier 5 is implemented in a semiconductor-on-insulator complementary metal oxide semiconductor (CMOS) process technology, or other CMOS process technology, because CMOS process technologies can have limited voltage handling capabilities as compared to other process technologies (e.g., Gallium Arsenide), and class F amplifiers can be well-suited under these circumstances. In other embodiments, other types of amplifiers such as class AB, inverted class F, or class E can be used.

Particularly where a class F amplifier is used, it can be desirable for the output matching circuit 9 to implement an effective second order harmonic short circuit and third order harmonic open circuit. Thus, the output matching circuits 9

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of FIGS. 2 and 3 can be configured to provide enough rejection on the second and third order harmonics.

Referring to FIGS. 2 and 3, the output of the power amplifier 5 and the input of the output matching circuit 9 are connected to the supply voltage biasing circuit 15. As shown in FIG. 3, the supply voltage biasing circuit 15 can be electrically connected between the power high supply voltage V_{CC} and the input node of the output matching network 9. The supply voltage biasing circuit 15 can be used to bias the power amplifier 5 with the power high supply voltage V_{CC} , which in certain implementations is generated by an envelope tracker or other supply control block. The supply voltage biasing circuit 15 includes an inductor 21 and a capacitor 23. The inductor can be referred to as a choke inductor 21 and includes a first end electrically connected to the power high supply voltage V_{CC} and a second end electrically connected to the drain of the first FET 17 at the output of the power amplifier 5. The choke inductor 21 can have an inductance sufficient to block RF signals generated by the power amplifier 5 from reaching the power high supply voltage V_{CC} . However, the choke inductor 21 should be sized to minimize $L \cdot dI/dt$ effects associated with receiver band noise, which can degrade performance in envelope tracking applications. The capacitor 23 can be referred to as a decoupling capacitor 23 and includes a first end electrically connected to the power high supply voltage V_{CC} and a second end electrically connected to the power low supply voltage V_L , and can perform a wide variety of functions. For example, including the decoupling capacitor 23 can reduce noise of the power high supply voltage V_{CC} and/or stabilize the output of the power amplifier 5. Additionally, the decoupling capacitor 23 can be used to provide an RF/AC ground to the second end of the choke inductor 21.

The choke inductor 21 in some embodiments is not implemented on the same die as the other components shown in FIG. 2. For instance, in one embodiment the choke inductor 21 is embedded in metal layers of a module laminate or substrate, as is further described with respect to FIG. 4. Such an embedded inductor can provide improved current handling. In another embodiment, the choke inductor 21 is implemented on the same die as other components illustrated in FIG. 2, such as the power amplifier, output matching network, and front end component 12. In yet another embodiment the choke inductor 21 is a surface mount component.

In various embodiments, the choke inductor 21 has an inductance in the range of about 1 nH to about 6 nH and the decoupling capacitor 23 has a capacitance in the range of about 0 nF (omitted) to about 5 nF. In one embodiment, the choke inductor 21 has an inductance of 1.1 nH and the decoupling capacitor 23 has a capacitance of 2.7 nF. However, other suitable inductance and capacitance values can be used, such as inductance and capacitance values associated with optimum load impedance and/or operating frequency.

The output matching network 9 includes first and second matching capacitors C1, C4, first and second second-order harmonic short circuits 25, 31, a third-order harmonic open circuit 27, a second order harmonic rejection circuit 29, and a DC blocking capacitor C7.

The first matching capacitor C1 and the second first second-order harmonic short circuit 25 are each electrically connected between the input node of the output matching network 9 and the power low supply voltage V_L . The third-order harmonic open circuit 27 is connected between the input node of the output matching network 9 and the first node 33. The second matching capacitor C4 is connected

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between the first node 33 and the power low supply voltage V_1 . The second order harmonic open circuit 29 is connected between the first node 33 and a second node 35. The second second-order harmonic short circuit 31 is connected between the second node 35 and the power supply low node V_1 . The DC blocking capacitor is connected between the second node 35 and the output node of the output matching network 9.

The first and second matching capacitors C1, C4 can be configured to match impedance of the output of the power amplifier 5.

The first and second second-order harmonic short circuits 25, 31 can be configured to resonate at about two times the fundamental frequency of the signal RF_IN so as to short second-order harmonic frequency signal components in the power amplifier's output signal. In the illustrated configuration, the first second-order harmonic short circuit 25 includes an inductor L1 and a capacitor C2 electrically connected in series between the power low supply voltage V_1 and the input node of the output matching network 9, which is also the output of the power amplifier 5 and, referring now only to FIG. 3, the drain of the first FET 17. The second second-order harmonic short circuit 31 includes an inductor L4 and a capacitor C6 electrically connected in series. Although the first and second second-order harmonic short circuits 25, 31 illustrate exemplary configurations, other configurations can be used, including, for example, implementations in which the order of the inductors and the capacitors in the series are reversed.

In one embodiment, the first second-order harmonic short circuit 25 is configured to resonate such that the impedance of the first second-order harmonic series resonant circuit 25 at two times the fundamental frequency of the signal RF_IN is less than about 1Ω . The first second-order harmonic short circuit 25 can be configured to resonate such that the impedance of the first second-order harmonic series resonant circuit 25 at two times the fundamental frequency of the signal RF_IN is less than about 20% of a load line impedance (e.g., a 5Ω load line impedance) of the output matching circuit 9.

The third-order harmonic open circuit 27 includes a capacitor C3 and an inductor L2 connected in parallel. The third-order harmonic open circuit 27 can be configured to resonate at about three times the fundamental frequency of the signal RF_IN and act as an open circuit, thereby blocking third-order harmonic frequency components generated by the output signal of the power amplifier 5 from reaching the first node 33.

The third-order harmonic open circuit 27 can improve the third-order harmonic rejection of the power amplifier 5 by providing high impedance to signals at about three times the fundamental frequency of the signal RF_IN. In one embodiment, the inductor L2 and the capacitor C3 are configured to resonate such that the impedance of the third-order harmonic parallel resonant circuit 27 at three times the fundamental frequency of the signal RF_IN is greater than about $50\text{ k}\Omega$. The inductor L2 and the capacitor C3 can be configured to resonate such that the impedance of the third-order harmonic open circuit 27 at three times the fundamental frequency of the signal RF_IN is greater than about 10000 times a load line impedance (e.g., a 5Ω load line impedance) of the output matching circuit 9.

The second order harmonic open circuit 29 includes a capacitor C5 and an inductor L3 connected in parallel. The second order harmonic open circuit 29 is configured to resonate at about two times the fundamental frequency of the signal RF_IN and act as an open circuit, thereby block-

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ing second-order harmonic frequency components generated by the output signal of the power amplifier 5 from reaching the second node 33. Thus, the second order harmonic open circuit 29 serves to supplement the second order harmonic rejection provided by the first and second second-order harmonic short circuits 25, 31.

The output matching circuit 9 further includes the DC blocking capacitor C7, which is electrically connected between the second node 35 and the output node of the output matching circuit 9. The DC blocking capacitor C7 can provide DC blocking and/or help provide an impedance transformation to achieve a desired load line impedance of the power amplifier 5 at the fundamental frequency. For example, in certain implementations the DC blocking capacitor C7 can be used at least in part to transform a termination impedance associated with the RF front end 10, such as a 50Ω termination impedance, to a load line impedance that is desirable for the power amplifier 5 from a power efficiency standpoint. Additionally, the DC blocking capacitor C7 can block DC signals, thereby helping to provide DC bias voltage isolation between the output of the power amplifier 5 and the input of the RF front end 10. Although FIGS. 2 and 3 illustrate a configuration including the DC blocking capacitor C7, in certain implementations the DC blocking capacitor C7 can be omitted, such as in implementations using a surface acoustic wave (SAW) filter.

Although the output matching circuit 9 has been illustrated in the context of one example of a power amplifier system, the output matching circuit 9 can be used in other configurations of power amplifier systems.

Referring now only to FIG. 3, the illustrated embodiment shows an implementation in which the capacitor C2 is implemented using a tunable bank of three switched capacitors C2a, C2b, C2c instead of the single capacitor C2 shown in the implementation of FIG. 2. The switches SW1 and SW0 are connected in series with the capacitors C2b and C2c respectively, such that the overall capacitance of C2 can be tuned. In particular, the overall capacitance can be tuned to one of three values: 1) C2a (where SW1 and SW0 are both off/open); 2) C2a+C2b (where SW1 is on/closed, SW0 is off/open); or 3) C2a+C2b+C2c (where SW1 and SW0 are both on/closed). In this manner, the front end 10 can dynamically adjust the output matching network to optimize the second order rejection provided by the first second-order short circuit 25.

While only the first second-order short circuit 25 is shown as having a dynamically adjustable capacitance value, any combination of one or more of the other capacitance or inductor values can be similar dynamically adjustable, e.g., by incorporating additional capacitors, inductors, and corresponding switches, as appropriate.

FIG. 3 shows the inductor L2 being implemented as an off-die component instead, not included on the same die as the other components of the output matching network 9. For instance, the inductor L2 can be a surface mount component mounted on a substrate of a module. Behavior of the output matching network 9 can be particularly sensitive to the precise value of L2, and a surface mount implementation of L2 can reduce the cost of selecting/adjusting the value of L2 as needed. While only L2 is shown as being a surface mount component in FIG. 3, any of the other capacitors or inductors can be surface mount components in other embodiments. In some embodiments, L2 is included on the same die (e.g., a semiconductor-on-insulator die) as the other components of the output matching network 9.

FIG. 4 depicts an example of a mobile device 400 including a radio frequency module 402 having a semicon-

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ductor-on-insulator die **404**. As shown, the semiconductor-on-insulator die can include a front end system **10**, which can be any of the front end systems described herein including those described previously with respect to FIGS. **1A-3**. The mobile device **400** can be an Internet of Things (IoT) capable device, for example. Additional details regarding compatible IoT networks and devices are provided herein, including in the section entitled Internet of Things Applications. The mobile device **400** of FIG. **4** can comprise or be incorporated into any of the IoT devices described in that section or elsewhere herein. In other embodiments, the mobile device **400** is a mobile phone

The front end system **10** includes one or more power amplifiers **5**, an output matching network **9**, one or more switches, and a controller **7**. The power amplifiers **5** can be similar to or the same as the similarly numbered power amplifiers in FIGS. **1-3**, and can in some embodiments be a cascode FET-based class F amplifier such as the power amplifier **5** of FIG. **3**. The output matching network **9** is configured to increase the power transfer and/or reduce reflections of the amplified RF signal generated by the power amplifier **5**, and can be any of the output matching networks described herein. The switch **12** can include one or more of the antenna-side switch **2** and transceiver-side switch **3** of FIGS. **1A-1B**, or the RF component **12** of FIGS. **2-3**. The controller **7** can be similar to or the same as the control and biasing circuit **7** of FIGS. **1A-1B**, and can be used to control and bias the components of the front-end system **10**. As one example, the controller **7** can include the biasing circuit **15** of FIGS. **2-3**, and can be configured to control the switches SW0, SW1 of FIG. **3** to tune the capacitance C2.

While not shown in FIG. **4**, the front end system **10** can include additional components, such as one or more LNAs, additional switches, bypass circuitry, or the like.

The module **402** can include one or more surface mount devices **406** mounted on the module substrate, and one or more embedded devices **408** embedded in metal layers of a laminate substrate of the module **402**. The surface mount devices **406** and embedded devices **408** can include inductors, capacitors, or other electrical components. For example, referring to FIG. **3**, the surface mount devices **406** can include the inductor L2, and the embedded devices **408** can include the choke inductor **21**.

In one embodiment, the module **400** implements at least a LTE-M communication standard and incorporates a CMOS silicon-on-insulator die **404**, eight surface mount devices **406** including the inductor L2, and one embedded device comprising the choke inductor L1.

While not shown, the module **406** can include additional components, including some or all of a transceiver, baseband processor, memory, antenna, and the like (see, e.g., FIGS. **13A-15**). Moreover, in some embodiments the mobile device **400** includes one or more additional modules or other components not shown in FIG. **4**, which can implement one or more of a transceiver, baseband processor, memory, antenna, or other appropriate componentry.

According to certain embodiments, the use of a semiconductor-on-insulator (e.g., silicon-on-insulator CMOS) process technology allows for the incorporation on a single die **404** multiple functional elements, whereas other previous solutions utilize multiple dies of different process technologies. Such single-die configurations can be particularly well-suited for Internet of Things applications and corresponding communication technologies.

The mobile devices and corresponding front end systems described herein can implement a variety of communication

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standards including LTE-M (LTE Machine Type Communication), Bluetooth, ZigBee, Z-Wave, 6LoWPAN, Thread, Wi-Fi, NFC, Sigfox, Neul, and/or LoRaWAN technologies. Furthermore, according to certain embodiments the mobile devices and corresponding front end systems can be configured to communicate using cellular infrastructure, for instance, using 2G, 3G, 4G (including LTE, LTE-Advanced, and/or LTE-Advanced Pro), and/or 5G technologies.

Semiconductor-on-Insulator Transistor Layout for Improved Heat Dissipation

Semiconductor-on-insulator process technologies such as silicon-on-insulator technologies include an insulator layer between the semiconductor wafer substrate and the active device region, e.g., to prevent device-to-substrate coupling loss. For example, any of the embodiments described herein that include semiconductor-on-insulator devices can include silicon wafers with a layer of silicon dioxide between the silicon substrate and the active device region, such as those shown and described with respect to FIGS. **4AA-4G**.

While the silicon substrate can have relatively good heat dissipation properties, the silicon dioxide layer typically does not, and can block heat dissipation through the wafer, making it impractical to use a wirebond package. Thus, according to certain embodiments the silicon-on-insulator die can be packaged in a flip chip bump pad package, where the bump contacts can assist with heat dissipation.

According to certain embodiments, semiconductor-on-insulator transistors are formed with a conduction path from the active device layer (e.g., from a source region of an active layer) to the ground plane in the module where at least portions of the conduction path are vertically contiguous. Such an arrangement can allow heat generated by the transistor to efficiently dissipate to the ground plane. For example, any of the power amplifiers described herein (e.g., the power amplifiers **5** of FIGS. **1-3**) can be formed in a semiconductor-on-insulator process technology and be implemented by transistors having such a layout.

FIG. **4A** is a cross-sectional view schematic of a portion of a semiconductor-on-insulator die **404** connected to a module **402**. FIG. **4AA** is a perspective view schematic of a semiconductor-on-insulator die **404** and module **402**. It will be appreciated that the layers can be etched or otherwise manipulated or processed during manufacture to create desired transistors, routing, and other electrical componentry. For example, FIGS. **4A** and **4AA** illustrate a transistor **420**, which can be the second FET transistor **19** of the power amplifier **5** of FIG. **3**, or form a part of the second FET transistor **19**. Moreover, while not shown, it will be appreciated that the die **404** and module **402** can include packages encapsulate and/or supporting the illustrated components, such as a package encapsulating each of the layers of the die **404** with the contact pad(s) **428** exposed on an underside of the package.

Referring to both FIGS. **4AA** and **4A**, the illustrated layers of the module **402** include the ground plane **422**, a module via layer **424**, and a module metal layer **426**, which can be a routing layer.

The illustrated layers of the semiconductor-on-insulator die **404** include a semiconductor wafer substrate **447**, an oxide layer **449**, an active device region **446** (also referred to herein as an "active layer"), a first via layer **444**, a first metal layer **442**, a second via layer **440**, a second metal layer **438**, a third via layer **436**, a third metal layer **434**, a fourth via layer **432**, a fourth metal layer **430**, and one or more bump contacts **428**. For example, the bump contact(s) **428** can be a copper pillar configured to physically contact the module routing layer **426** and electrically connect the die

404 to the module 402 and the ground plane 422. The fourth metal layer 430 is connected to and above the bump contact 428. The fourth via layer 432 connects the fourth metal layer 430 to the third metal layer 434. The third via layer 436 connects the third metal layer 434 to the second metal layer 438. The second via layer 440 connects the second metal layer 438 to the first metal layer 442. The first via layer 444 connects the first metal layer 442 to the portion of the active device region 446 corresponding to the source 448 of the transistor 420.

The transistor 420 (FIG. 4A) includes a source 448, drain 450, and body 452 formed in the active device region 446, as well as a gate 454 formed in a polysilicon layer.

The oxide layer 449 is formed between the active device region 446 and the wafer substrate 447. For example, in some embodiments the wafer substrate 450 is formed from silicon, the oxide layer 449 is silicon dioxide, and the active device region 446 comprises doped silicon.

As represented by the dashed line 451, an electrical coupling path between the source 448 of the transistor 420 and the bump contact 428 can include an at least partially direct vertical physical path to the ground plane 422 of the module 402. Such a configuration can allow for enhanced dissipation of heat generated by the transistor 420. For example, in some embodiments at least some portion of the conduction path extends along the entirety of the portion of the line 451 between the source terminal 458 of the transistor 420 to the ground plane 422. The line 451 is orthogonal to a plane P, where the plane P is substantially parallel, or is parallel, to planes defined by top and bottom planar surfaces of the die 404 and its constituent layers and/or the module 402 and its constituent layers.

In order to provide for the substantially contiguous vertical conduction path, or an at least partially substantially contiguous vertical conduction path, according to certain embodiments, there can be at least some surface area overlap between each adjacent layer in the conduction path from the source 448 to the ground plane 422.

According to some embodiments, the surface area of the interface between the first via layer 444 and the source 448 can be at least 4, 5, 6, 7, or 8 percent of the surface area of the source 448. For example, according to certain embodiments the surface area of the interface between the first via layer 444 and the source 448 is 7.2 percent of the surface area of the source 448. Moreover, in some implementations, the source 448 can substantially completely overlap, or completely overlap, the first via layer 444.

In certain embodiments, substantially the entirety of the portion of the source conduction path that includes the active device region layer 446, the first via layer 444, the first metal layer 442, the second via layer 440, and the second metal layer 438 extends substantially orthogonal, or orthogonal, to the plane P. The surface area of an interface in the source conduction path between the second via layer 440 and the first metal layer 442 can be at least 10, 15, 20, 25, 26, or 27 percent of the surface area of the portion of the first metal layer 442 that resides in the source conduction path. For example, the surface area of the interface in the source conduction path between the first second via layer 440 and the first metal layer can be at least 27 percent of the surface area of the portion of the first metal layer 442 that resides in the source conduction path.

According to certain embodiments, some or all of the conduction path from the transistor 420 to the bump contact 428 or to the ground plane 422 comprises a contiguous vertical path, e.g., orthogonal to the plane P, without including horizontal vias or other horizontally routed connections.

This can be advantageous because horizontal connections can be relatively thin and therefore not good for heat dissipation as compared to vertical conduction. As will be appreciated, FIGS. 4A-4AA are simplified schematic views intended to illustrate the arrangement of the various components of the die 404 and module 402 with respect to one another. Moreover, while FIGS. 4A-4AA illustrate a single source connection path for a single transistor 420, it will be appreciated that additional components can be implemented in the die 404, and additional connection paths exist for the transistor 420 that are not illustrated, such as connection paths corresponding to the drain 450 and gate 454. Moreover, while various components are described as "layers" with respect to FIGS. 4A and 4AA, it will be appreciated that some or all of the illustrated portions (e.g., 424, 426, 428, 430, 432, 434, 436, 438, 440, 442, 444, 446, 454) can actually correspond to etched or otherwise partial segments of larger layers.

FIG. 4B is a top view showing certain layers of a portion of a semiconductor-on-insulator die 404, depicting first and second transistors 420a, 420b of a cascode power amplifier implemented on the die 404. For example, the first and second transistors 420a, 420b can be the first and second FETs 17, 19 of the power amplifier 5 of FIG. 3.

The first transistor 420a includes a first half 460 and a second half 462. The second transistor 19 includes a first half 464 and a second half 466.

Referring now to the first half 464 of the second transistor 420b for the purposes of illustration, a first oval corresponds to an outline of the module metal layer 426, and an octagon corresponds to an outline of a contact bump 428 corresponding to the first half of the second transistor 420b. As shown, the octagonal footprint of the contact bump 428 overlaps with the footprint of the second transistor 420b. In particular, the footprint of the contact bump 428 substantially overlaps with the footprint of the first half 464 of the second transistor 420b. Similarly, the octagonal footprint of a second contact bump overlaps with the footprint of the second half 466 of the second transistor 420b.

FIG. 4C is a zoomed in view of the first half 464 of the second transistor 420b. FIG. 4D is a further zoomed in view showing portions of first half 464 of the second transistor 420b. In FIG. 4D, first portions 442g of the first metal layer 442 are called out. The first portions 442g of the first metal layer 442 correspond to gate contacts, connected to the gate of the second transistor 420b. Thus, the first portions 442g do not correspond to the portion of the first metal layer 442 shown in FIG. 4A, and do not reside in the source conduction path. The module via layer 426 is also illustrated in FIG. 4C (boxes with x's). FIG. 4E is a further zoomed in view showing portions of the first half 464 of the second transistor 420b. In FIG. 4E, the following additional layers are called out: the fourth metal layer 430 (bounded by rectangle), the fourth via layer 432 (larger 'x's), the third metal layer 434 (interior rectangular outlines), the third via layer 436 (smaller 'x's), and the second metal layer 438 (bounded by rectangle).

FIG. 4F shows certain layers of the portion 468 of the first half 464 of the second transistor 420b outlined by the dashed box in FIG. 4E. FIG. 4F further calls out portions of the polysilicon layer 454 (rectangular strips) that form portions of the gate of the second transistor 420b, portions of the active device region 446 (bounded by rectangles), second and third portions 442s, 442d of the first metal layer 442 (rectangular strips), and portions of the first via layer 444 (squares with x's overlapping metal layer portions 442s, 442d). The second portions 442s of the first metal layer 442

correspond to portions of the first metal layer **442** contacting the source **448** of the second transistor **420** and the third portions **442d** of the first metal layer **442** correspond to portions of the first metal layer **442** contacting the drain of the second transistor **420b**. Thus, the portions **442s** correspond to and are included in the source conduction path **451**.

FIG. 4F shows that a leftmost strip **446s1** of the portions **446s** (see also FIG. 4G) of the active region layer **446** corresponding to the left most source region of the portion **468** of the second transistor **420b** has a width *W* and a length *L*. According to one embodiment, the length *L* is 10 microns, the width *W* is 0.5 microns. In this embodiment, there are 9 active square portions of the first via layer **444** that overlap the strip **446s1**, and which form a part of the source conduction path. The remaining illustrated square portions are not active or are otherwise not connected in the source conduction path. Each of the 9 squares included in the source conduction path is 0.2 microns by 0.2 microns. Thus, the portions of the first via layer **444** overlapping and interfacing with the leftmost strip **446s1** overlap and interface with 7.2 percent of the leftmost strip **446s1** according to the following calculation: $(0.2 \text{ microns} \times 0.2 \text{ microns} \times 9) / (10 \text{ microns} \times 0.5 \text{ microns})$. Because each of the remaining strips of active source region **446s** have the same or similar dimensions *L*, *W*, and also each overlap with 9 active 0.2x0.2 micron square portions of the first via layer **444**, the 7.2 percentage can be extrapolated across the entire source conduction path of the second transistor **420b**. In other words, in this example embodiment, the portions of the first via layer **444** corresponding to the source **448** conduction path of the transistor **420b** overlap with and are in contact with 7.2 percent of the surface area of the portion of the active region **446** corresponding to the source conduction path. As such, the surface area of the interface between the source **448** of the transistor **420b** and the first via layer **444** is 7.2 percent of the surface area of the overall surface area of the source **448**. According to various embodiments, the surface area of the interface between the first via layer **444** and the source **448** can be at least 4, 5, 6, 7, or 8 percent of the surface area of the source **448**, which can provide improved heat dissipation, among other benefits.

FIG. 4G further calls out, for the zoomed-in portion **468** of the transistor **420b**, portions of the second via layer **440** (squares with cross-bars) as well as the portions **446s** (three rectangular strips) of the active layer **446** corresponding to the source **448** and the portions **446d** (three rectangular strips) of the active layer **446** corresponding to the drain **450**. In the illustrated embodiment, each strip of the first metal layer **442s** corresponding to the source conduction path is overlaid by 8 square portions of the second via layer **440**. In some embodiments, the combined surface area of the 8 square portions is 27 percent of the surface area of the corresponding strip **442s**. Because each of the strips **442s** corresponding to the transistor **420b** have the same or similar dimensions, and each have 8 similar square portions of the second via layer **440**, the 27 percent can be extrapolated across the entire source conduction path of the second transistor **420b**. In other words, in this example embodiment, the portions of the second via layer **444** corresponding to the source **448** conduction path of the second transistor **420b** overlap with and are in contact with 27 percent of the surface area of the portion of first metal layer **442** in the source conduction path. As such, the surface area of the interface between the second via layer **444** and the first metal layer **442** for the source conduction path of the second transistor **420b** is 27 percent. This significant overlap can provide improved heat dissipation, among other benefits. In

various embodiments, the surface area of the interface in the source conduction path between the second via layer **440** and the first metal layer **442** can be at least 10, 15, 20, 25, 26, or 27 percent of the surface area of the portion of the first metal layer **442** that resides in the source conduction path.

At least some portion of the conduction path can extend along the entirety of a line **451** that extends from the source terminal **458** of the transistor **420** to the contact pad **428**, from the source terminal **458** to the ground plane **422**, or along the entirety of any portion of that line **451**, where the line **451** is orthogonal to a plane *P* (FIG. 4A). For example, referring to FIGS. 4B-4G, according to certain embodiments such portions of the source conduction path fall within the illustrated footprints of all of the following layers, thereby creating a straight line connection: active region layer **446s** (FIGS. 4F and 4G), first via layer **444** (FIG. 4F), first metal layer **442s** (FIGS. 4F and 4G), second via layer **440** (FIG. 4G), second metal layer **438** (FIG. 4E), third via layer **436** (FIG. 4E), third metal layer **434** (FIG. 4E), fourth via layer **432** (FIG. 4E), fourth metal layer **430** (FIG. 4E), contact pad **428** (FIGS. 4B and 4C), and module layer **426** (FIGS. 4B and 4C). Such vertical routing can provide improved heat dissipation during transistor operation.

Moreover, the conduction path extending from the contact pad **428** to a terminal (e.g., source **448** or drain **450**) of the transistors **420a**, **420b** can include a plurality of first conduction path portions that reside entirely within the footprints of the contact pad(s) **428** connected to the respective transistor **420a**, **420b** when viewed from above the die. For example, the first half **464** of the second transistor **420b** includes an active portion **465** (FIG. 4C) that includes the source/drain portions **446s/446d** (FIGS. 4F-4G) and/or the corresponding portions **442s/442d** (FIGS. 4F-4G) of the first metal layer **442**. As shown, the majority of the active portion **465** of the first half **464** of the second transistor **420b** falls within the footprint of and is overlapped by the upper left contact pad **428**. Likewise, the majority of the active portion of the second half **466** of the second transistor **420b** resides within the footprint of the lower left contact pad **428**. Thus, given the generally vertical routing scheme between the source/drain portions **446s/446d** and the contact pads **428**, and given the repeating cell structure of the transistor **420b**, the conduction path will include first conduction path portions residing within the footprints of and overlapped by the two contact pads **428**. Moreover, in the illustrated embodiment, because the majority of the active portion **465** is within the footprints of the two contact pads **428**, the first conduction path portions make up a majority of the overall conduction path between the terminals (e.g., source, drain) of the second transistor **420b** and the two contact pads **428**. With reference to the source terminal **448**, for example, the first conduction path portions generally include the portions of the following layers that form electrically connected pathways falling entirely within the footprints of the two contact pads **428** and extending from the active region layer **446s** to the contact pads **428**: active region layer **446s** (FIGS. 4F and 4G), first via layer **444** (FIG. 4F), first metal layer **442s** (FIGS. 4F and 4G), second via layer **440** (FIG. 4G), second metal layer **438** (FIG. 4E), third via layer **436** (FIG. 4E), third metal layer **434** (FIG. 4E), fourth via layer **432** (FIG. 4E), and fourth metal layer **430** (FIG. 4E). Conversely, the remainder of the conduction path is generally comprised of second conductive path portions corresponding to the active portions of the first and second halves **464**, **466** of the second transistor **420b** that fall outside of and are not overlapped by the footprints of the two contact pads **428**. Depending on the embodiment, at least 25, 30, 40, 45, 50,

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55, or 60 percent of the conduction path between the terminals of each transistor **420a**, **420b** resides entirely within the footprint of the two contact pads **428** corresponding to the transistor **420a**, **420b**. Such significant overlap between contact pads and transistor terminals can lead to improved heat dissipation.

Referring to FIGS. **4B**, **4F**, and **4G**, at least some of the active layer portions **446s** of the source **448** can fall within a footprint of the at least one contact pad **428** of the second transistor **420b**. For example, in the illustrated example more than 50% of the active layer portions **446s** of the source **448** corresponding to the first half **464** of the second transistor **420b** fall within the octagonal footprint of the top contact pad **428**. Similarly, more than 50% of the active layer portions **446s** of the source **448** corresponding to the second half **466** of the second transistor **420b** fall within the octagonal footprint of the bottom contact pad **428**. Thus, more than 50% of the active layer portions **446s** of the source **448** across the entire transistor **420b** fall within the footprint of the two contact pads that form the at least one contact pad **428**, which can provide improved heat dissipation. In various embodiments, at least about 25, 30, 40, 45, 50, 55, or 60 percent of the source **448** can fall within the footprint of the at least one contact pad **428**.

Referring still to FIGS. **4B**, **4F**, and **4G**, at least some of the source **448** can fall within a footprint of the module metal layer **426** corresponding to the source conduction path. For instance, substantially the entire source **448** can fall within a footprint of the module metal layer **426** corresponding to the source conduction path, such as in the illustrated example where all of the active layer portions **446s** of the source **448** of both the first and second halves **464**, **466** of the transistor **420b** fall within the footprint of the first and second portions of the module metal layer **426** delineated by the two called-out oval outlines. In various embodiments, at least about 50, 75, 80, 90 or 100 percent of the source **448** can fall within the footprint of the portion of the module metal layer **426** corresponding to the source conduction path, providing improved heat dissipation.

While certain aspects have been described with respect to the second transistor **420b** for the purposes of illustration, given the generally symmetrical nature of the implementation of the first and second transistors **420a**, **420b**, it will be appreciated that similar properties apply to the first transistor **420a**.

Packaged Modules

FIG. **5A** is a schematic diagram of one embodiment of a packaged module **900** which can incorporate any of the front end systems described herein or otherwise be incorporated into any of the mobile devices included herein. For example, the module **900** may correspond to the module **400** of FIG. **4**. FIG. **5B** is a schematic diagram of a cross-section of the packaged module **900** of FIG. **5A** taken along the lines **5B-5B**.

The packaged module **900** includes radio frequency components **901**, a semiconductor die **902**, surface mount devices **903**, wirebonds **908**, a package substrate **920**, and an encapsulation structure **940**. For example, the semiconductor die **902** can be the semiconductor die **404** of FIG. **4**, and can incorporate the front end systems **10** of FIG. **1A**, **2**, or **3**, the front end system **20** of FIG. **1B**. The package substrate **920** includes pads **906** formed from conductors disposed therein. Additionally, the semiconductor die **902** includes pins or pads **904**, and the wirebonds **908** have been used to connect the pads **904** of the die **902** to the pads **906** of the package substrate **920**.

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The semiconductor die **902** includes a power amplifier **945**, which can be implemented in accordance with one or more features disclosed herein. While only the power amplifier **945** is shown for simplicity, it will be appreciated that additional componentry including any of the output matching networks, LNAs, switches, controllers, and the like can be included.

The packaging substrate **920** can be configured to receive a plurality of components such as radio frequency components **901**, the semiconductor die **902** and the surface mount devices **903**, which can include, for example, surface mount capacitors and/or inductors. In one implementation, the radio frequency components **901** include integrated passive devices (IPDs).

As shown in FIG. **5B**, the packaged module **900** is shown to include a plurality of contact pads **932** disposed on the side of the packaged module **900** opposite the side used to mount the semiconductor die **902**. Configuring the packaged module **900** in this manner can aid in connecting the packaged module **900** to a circuit board, such as a phone board of a mobile device. The example contact pads **932** can be configured to provide radio frequency signals, bias signals, and/or power (for example, a power supply voltage and ground) to the semiconductor die **902** and/or other components. As shown in FIG. **5B**, the electrical connections between the contact pads **932** and the semiconductor die **902** can be facilitated by connections **933** through the package substrate **920**. The connections **933** can represent electrical paths formed through the package substrate **920**, such as connections associated with vias and conductors of a multilayer laminated package substrate.

In some embodiments, the packaged module **900** can also include one or more packaging structures to, for example, provide protection and/or facilitate handling. Such a packaging structure can include overmold or encapsulation structure **940** formed over the packaging substrate **920** and the components and die(s) disposed thereon.

It will be understood that although the packaged module **900** is described in the context of electrical connections based on wirebonds, one or more features of the present disclosure can also be implemented in other packaging configurations, including, for example, flip-chip configurations.

Internet of Things Applications

As mentioned previously, one example application of the front end systems herein is to enable various objects with wireless connectivity, such as for Internet of things (IoT). IoT refers to a network of objects or things, such as devices, vehicles, and/or other items that are embedded with electronics that enable the objects to collect and exchange data (for instance, machine-to-machine communications) and/or to be remotely sensed and/or controlled. The front end systems herein can be used to enable wireless connectivity of various objects, thereby allowing such objects to communicate in an IoT network. The front end systems discussed herein can be implemented in IoT applications to enable wireless connectivity to expand the way consumers manage information and their environment. Such front end systems can enable the new and emerging IoT applications, which can bring people and things closer to vital information wherever it is desired. Although IoT is one example application of front end systems herein, the teachings herein are applicable to a wide range of technologies and applications. Some example IoT applications will now be discussed.

IoT devices can be implemented in automotive systems. From telematics to infotainment systems, lighting, remote keyless entry, collision avoidance platforms, toll transpon-

ders, video displays, vehicle tracking tools, and the like, front end systems in accordance with any suitable principles and advantages discussed herein can help enable convenience and safety features for the connected vehicle.

IoT devices can be implemented in connected home environments. Front end systems in accordance with any suitable principles and advantages discussed herein can allow homeowners greater control over their home environment. IoT devices can be implemented in a host of devices including smart thermostats, security systems, sensors, light switches, smoke and carbon monoxide alarms, routers, high definition televisions, gaming consoles and much more.

IoT devices can be implemented in industrial contexts. From smart city applications to factory automation, building controls, commercial aircraft, vehicle tracking, smart metering, LED lighting, security cameras, and smart agriculture functions, front ends systems in accordance with any suitable principles and advantages discussed herein can enable these applications and meet specifications.

IoT devices can be implemented in machine-to-machine contexts. IoT devices can enable machine-to-machine communications that can transform the way organizations do business. From manufacturing automation to telemetry, remote control devices, and asset management, front end systems discussed herein can provide cellular, short-range, and global positioning solutions that support a wide range of machine-to-machine applications.

IoT devices can be implemented in medical applications. Front end systems in accordance with any suitable principles and advantages discussed herein can enable medical devices and the communication of information that is improving the care of millions of people worldwide. Front end systems in accordance with any suitable principles and advantages discussed herein can be integrated into product designs that enable the miniaturization of medical devices and enhance data transmission. Amplifiers, such as power amplifiers and low noise amplifiers, in accordance with any suitable principles and advantages discussed herein can be implemented in medical instruments.

IoT devices can be implemented in mobile devices. The communication landscape has changed in recent years as consumers increasingly seek to be connected everywhere and all the time. Front end systems in accordance with any suitable principles and advantages discussed herein can be compact, energy and cost efficient, meeting size and performance constraints, while enabling a great consumer experience. Wireless mobile devices, such as smartphones, tablets and WLAN systems, can include a front end system in accordance with any suitable principles and advantages discussed herein.

IoT devices can be implemented in smart energy applications. Utility companies are modernizing their systems using computer-based remote control and automation that involves two-way communication. Some benefits to utilities and consumers include optimized energy efficiency, leveling and load balancing on the smart grid. Front end systems in accordance with any suitable principles and advantages discussed herein can be implemented in smart meters, smart thermostats, in-home displays, LTE-M (LTE Machine Type Communication), ZigBee/802.15.4, Bluetooth, and Bluetooth low energy applications.

IoT devices can be implemented in wearable devices. Wearable devices, such as smartwatches, smart eyewear, fitness trackers and health monitors, can include front end systems in accordance with any suitable principles and advantages discussed herein to enable relatively small form factor solutions that consume relatively low power and

enable always on connectivity. This can allow applications to run in the background for lengthy periods of time without a battery recharge, for example.

Any suitable principles and advantages discussed herein can be implemented in an IoT network, IoT object, a vehicle, industrial equipment, a corresponding front end system, a corresponding circuit board, the like, or any suitable combination thereof. Some examples will now be discussed.

FIG. 6 is a schematic diagram of one example of an IoT network 200. The IoT network 200 includes a smart home 201, a smart vehicle 202, a wearable 203, a mobile device 204, a base station 205, a smart hospital 206, a smart factory 207, and a smart satellite 208. One or more of the IoT-enabled objects of FIG. 6 can include a front end system, such as a front end module and/or front-end integrated circuit, implemented in accordance with the teachings herein.

The smart home 201 is depicted as including a wide variety of IoT-enabled objects, including an IoT-enabled router 211, an IoT-enabled thermostat 212, an IoT-enabled meter 213, IoT-enabled laptop 214, and an IoT-enabled television 215. Although various examples of IoT-enabled objects for a smart home are shown, a smart home can include a wide variety of IoT-enabled objects. Examples of such IoT-enabled objects include, but are not limited to, an IoT-enabled computer, an IoT-enabled laptop, an IoT-enabled tablet, an IoT-enabled computer monitor, an IoT-enabled television, an IoT-enabled media system, an IoT-enabled gaming system, an IoT-enabled camcorder, an IoT-enabled camera, an IoT-enabled modem, an IoT-enabled router, an IoT-enabled kitchen appliance, an IoT-enabled telephone, an IoT-enabled air conditioner, an IoT-enabled washer, an IoT-enabled dryer, an IoT-enabled copier, an IoT-enabled facsimile machine, an IoT-enabled scanner, an IoT-enabled printer, an IoT-enabled scale, an IoT-enabled home assistant (for instance, a voice-controlled assistant device), an IoT-enabled security system, an IoT-enabled thermostat, an IoT-enabled smoke detector, an IoT-enabled garage door, an IoT-enabled lock, an IoT-enabled sprinkler, an IoT-enabled water heater, and/or an IoT-enabled light.

As shown in FIG. 6, the smart vehicle 202 also operates in the IoT network 200. The smart vehicle 202 can include a wide variety of IoT-enabled objects, including, but not limited to, an IoT-enabled infotainment system, an IoT-enabled lighting system, an IoT-enabled temperature control system, an IoT-enabled lock, an IoT-enabled ignition, an IoT-enabled collision avoidance system, an IoT-enabled toll transponder, and/or an IoT-enabled vehicle tracking system. In certain implementations, the smart vehicle 202 can communicate with other smart vehicles to thereby provide vehicle-to-vehicle (V2V) communications. Furthermore, in certain implementations the smart vehicle 202 can operate using vehicle-to-everything (V2X) communications, thereby communicating with traffic lights, toll gates, and/or other IoT-enabled objects.

The wearable 203 of FIG. 6 is also IoT-enabled. Examples of IoT-enabled wearables include, but are not limited to, an IoT-enabled watch, an IoT-enabled eyewear, an IoT-enabled fitness tracker, and/or an IoT-enabled biometric device.

The IoT network 200 also includes the mobile device 204 and base station 205. Thus, in certain implementations user equipment (UE) and/or base stations of a cellular network can operate in an IoT network and be IoT-enabled. Furthermore, a wide variety of IoT-enabled objects can communicate using existing network infrastructure, such as cellular infrastructure.

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With continuing reference to FIG. 6, IoT is not only applicable to consumer devices and objects, but also to other applications, such as medical, commercial, industrial, aerospace, and/or defense applications. For example, the smart hospital **206** can include a wide variety of IoT-enabled medical equipment and/or the smart factory **207** can include a wide variety of IoT-enabled industrial equipment. Furthermore, airplanes, satellites, and/or aerospace equipment can also be connected to an IoT network. Other examples of IoT applications include, but are not limited to, asset tracking, fleet management, digital signage, smart vending, environmental monitoring, city infrastructure (for instance, smart street lighting), toll collection, and/or point-of-sale.

Although various examples of IoT-enabled objects are illustrated in FIG. 6, an IoT network can include a wide variety of types of objects. Furthermore, any number of such objects can be present in an IoT network. For instance, an IoT network can include millions or billions of IoT-enabled objects or things.

IoT-enabled objects can communicate using a wide variety of communication technologies, including, but not limited to, LTE-M (LTE Machine Type Communication), Bluetooth, ZigBee, Z-Wave, 6LoWPAN, Thread, Wi-Fi, NFC, Sigfox, Neul, and/or LoRaWAN technologies. Furthermore, certain IoT-enabled objects can communicate using cellular infrastructure, for instance, using 2G, 3G, 4G (including LTE, LTE-Advanced, and/or LTE-Advanced Pro), and/or 5G technologies.

FIG. 7A is a schematic diagram of one example of an IoT-enabled watch **300**. The IoT-enabled watch **300** illustrates one example of a smart wearable that can include a front end system implemented in accordance with one or more features disclosed herein.

FIG. 7B is a schematic diagram of one example of a front end system **301** for an IoT-enabled object, such as the IoT-enabled watch **300** of FIG. 7A. The front end system **301** includes a first transceiver-side switch **303**, a second transceiver-side switch **304**, a first antenna-side switch **305**, a second antenna-side switch **306**, a first power amplifier **307**, a second power amplifier **308**, a duplexer **311**, a directional coupler **312**, a termination impedance **313**, a first band selection filter **315**, a second band selection filter **316**, and a third band selection filter **317**. While not shown, the front end system **301** can further include any of the output matching networks described herein.

In the illustrated embodiment, the first transceiver-side switch **303** selects between a Band 26 transmit input pin (B26 TX IN) and a Band 13 transmit input pin (B13 TX IN). The second transceiver-side switch **303** controls connection of the output of the first power amplifier **307** to the first band selection filter **315** or the first band selection filter **316**. Thus, the first power amplifier **307** selectively amplifies Band 26 or Band 13, in this example. Additionally, the second power amplifier **308** amplifies a Band 12 transmit input pin (B12 TX IN). After suitable filtering by the band selection filters **315-317**, the second antenna-side switch **306** selects a desired transmit signal for providing to an antenna pin (ANT) via the duplexer **311** and the directional coupler **312**. As shown in FIG. 7B, the directional coupler **312** is terminated by the termination impedance **313**. Additionally, the first antenna-side switch **305** provides a signal received on the antenna pin (ANT) to a desired receive output pin (four in this example) of the front end system **301**. The illustrated front end system **301** also includes various additional pins to provide additional functionality, such as enhanced monitoring of transmit power. For instance, front end system **301** includes a directional coupler output pin (CPL), and feed-

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back pins (B12 RX, B13 RX, and B26 RX) for providing feedback signals associated with transmit signals (for Band 12, Band 13, and Band 26, respectively) generated by the power amplifiers.

The front end system **301** can incorporate one or more features described in the sections herein.

FIG. 8A is a schematic diagram of one example of IoT-enabled vehicles **321a-321d**. Each of the IoT-enabled vehicles **321a-321d** includes a front end system for enabling wireless vehicle-to-vehicle communications. The IoT-enabled vehicles **321a-321d** can include a front end system implemented in accordance with one or more features disclosed herein.

FIG. 8B is a schematic diagram of another example of a front end system **325** for an IoT-enabled object. The front end system **325** includes an antenna-side switch **331**, a bypass switch **332**, an LNA **333**, and a bias and logic circuit **334**.

The front end system **325** includes control pins (C0 and C1) for controlling the front end system **325** and a supply voltage pin (VDD) for powering the front end system **325**. The antenna-side switch **331** selectively connects an antenna pin (ANT) to a transmit signal pin (TX_IN) or a receive signal pin (RX_OUT). The LNA **333** includes an input connected to an LNA input pin (LNA_IN) and an output connected to the LNA output pin (LNA_OUT). The LNA **333** is selectively bypassed by the bypass switch **332**. Using external conductors and components, the LNA input pin (LNA_IN) can be connected to the receive signal pin (RX_OUT) either directly or indirectly (for instance, via a filter or other components). Furthermore, an external power amplifier can provide a transmit signal to the transmit signal pin (TX_IN).

The front end system **325** can incorporate one or more features described in the sections herein.

FIG. 9A is a schematic diagram of one example of IoT-enabled industrial equipment **340**. In the illustrated embodiment, the IoT-enabled industrial equipment **340** includes heliostats **341** for reflecting light to a solar receiver and turbine **342**. The IoT-enabled industrial equipment **340** can include one or more front end systems for a variety of purposes, such as providing angular positional control of the heliostats **341** to control concentration of solar energy directed toward the solar receiver and turbine **342**. The IoT-enabled industrial equipment **340** can include a front end system implemented in accordance with one or more features disclosed herein.

FIG. 9B is a schematic diagram of another example of a front end system **345** for an IoT-enabled object, such as the IoT-enabled industrial equipment **340** of FIG. 9A.

The front end system **345** includes a logic control circuit **350**, a transceiver DC blocking capacitor **351**, a first antenna DC blocking capacitor **352**, a second antenna DC blocking capacitor **353**, an LNA **354**, a power amplifier **356**, an antenna-side switch **357**, a bypass switch **358**, and a transceiver-side switch **359**. While not shown, the front end system **301** can further include any of the output matching networks described herein.

The front end system **345** includes control pins (CPS, CTX, CSD, ANT_SEL) for controlling the front end system **345**. The antenna-side switch **357** selectively connects either a first antenna pin (ANT1) or a second antenna pin (ANT2) to either an output of the power amplifier **356** or the bypass switch **358**/input to the LNA **354**. Additionally, the bypass switch **358** selectively bypasses the LNA **354**. Furthermore, the transceiver-side switch **359** selectively connected the transceiver pin (TR) to either an input of the power amplifier

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356 or the bypass switch 358/output to the LNA 354. The DC blocking capacitors 351-353 serve to provide DC blocking to provide enhanced flexibility in controlling internal DC biasing of the front end system 345.

The front end system 345 can incorporate one or more features described in the sections herein.

FIG. 10A is a schematic diagram of one example of an IoT-enabled lock 360. The IoT-enabled lock 360 illustrates one example of an IoT-enabled object that can include a front end system implemented in accordance with one or more features disclosed herein.

FIG. 10B is a schematic diagram of one example of a circuit board 361 for the IoT-enabled lock 360 of FIG. 10A. The circuit board 361 includes a front end system 362, which can incorporate one or more features described in the sections herein.

FIG. 11A is a schematic diagram of one example of IoT-enabled thermostat 370. The IoT-enabled thermostat 370 illustrates another example of an IoT-enabled object that can include a front end system implemented in accordance with one or more features disclosed herein.

FIG. 11B is a schematic diagram of one example of a circuit board 371 for the IoT-enabled thermostat 370 of FIG. 11A. The circuit board 371 includes a front end system 372, which can incorporate one or more features described in the sections herein.

FIG. 12A is a schematic diagram of one example of IoT-enabled light 380. The IoT-enabled light 380 illustrates another example of an IoT-enabled object that can include a front end system implemented in accordance with one or more features disclosed herein.

FIG. 12B is a schematic diagram of one example of a circuit board 381 for the IoT-enabled light 380 of FIG. 12A. FIG. 12B also depicts a base portion of the IoT-enabled light 380 for housing the circuit board 381. The circuit board 381 includes a front end system 382, which can incorporate one or more features described in the sections herein.

Radio Frequency Systems

FIG. 13A-13F illustrates various schematic block diagrams of examples of radio frequency systems that include a front end system, such as a front end module or front end integrated circuit. The radio frequency systems of FIGS. 13A-13F can incorporate one or more features described in the sections herein. In certain implementations, a radio frequency system, such as any of the radio frequency systems of FIGS. 13A-13F, is implemented on a circuit board (for instance, a printed circuit board (PCB)) of a wireless communication device, such as a mobile phone, a tablet, a base station, a network access point, customer-premises equipment (CPE), an IoT-enabled object, a laptop, and/or a wearable electronic device.

FIG. 13A illustrates a schematic block diagram of one example of a radio frequency system 500. The radio frequency system 500 includes an antenna 501, a front end system 10, and a transceiver 505. As was discussed above, the front end system 10 can incorporate one or more features described in the sections herein.

The antenna 501 operates to wirelessly transmit RF signals received via the antenna-side switch 2. The RF transmit signals can include RF signals generated by the power amplifier 5 and/or RF signals sent via the bypass circuit 4. The output matching network 9 receives the signal generated by the power amplifier 5. The antenna 501 also operates to wirelessly receive RF signals, which can be provided to the LNA 6 and/or the bypass circuit 4 via the antenna-side switch 2. Although an example where a common antenna is used for transmitting and receiving signals,

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the teachings herein are also applicable to implementations using separate antennas for transmission and reception. Example implementations of the antenna 501 include, but are not limited to, a patch antenna, a dipole antenna, a ceramic resonator, a stamped metal antenna, a laser direct structuring antenna, and/or a multi-layered antenna.

The transceiver 505 operates to provide RF signals to the transceiver-side switch 3 for transmission and/or to receive RF signals from the transceiver-side switch 3. The transceiver 505 can communicate using a wide variety of communication technologies, including, but not limited to, one or more of 2G, 3G, 4G (including LTE, LTE-Advanced, and/or LTE-Advanced Pro), 5G, WLAN (for instance, Wi-Fi), WPAN (for instance, LTE-M, Bluetooth and/or ZigBee), WMAN (for instance, WiMAX), and/or GPS technologies.

FIG. 13B illustrates a schematic block diagram of another example of a radio frequency system 506. The radio frequency system 506 includes a front end system 20 and a transceiver 505. As was discussed above, the front end system 20 can incorporate one or more features described in the sections herein.

FIG. 13C illustrates a schematic block diagram of another example of a radio frequency system 510. The radio frequency system 510 includes an antenna 501, a front end system 511, and a transceiver 505. The front end system 511 of FIG. 13C is similar to the front end system 10 of FIG. 13A, except that the bypass path including the bypass circuit 4 has been omitted and the antenna-side switch 2' and the transceiver-side switch 3' include one less throw. Thus, the antenna-side switch 2' is configured to selectively electrically connect the antenna 501 to either an input to the LNA 6 or an output of the power amplifier 5. Additionally, the transceiver-side switch 3' is configured to selectively electrically connect the transceiver 505 to either an output to the LNA 6 or an input of the power amplifier 5.

FIG. 13D illustrates a schematic block diagram of another example of a radio frequency system 512. The radio frequency system 512 includes a first antenna 501, a second antenna 502, a front end system 514, and a transceiver 505. The front end system 514 of FIG. 13D is similar to the front end system 10 of FIG. 13A, except that the antenna-side switch 2" includes an additional throw to provide connectivity to an additional antenna. Thus, the bypass circuit 4, the power amplifier 5, and/or the LNA 6 can be selectively electrically connected to the first antenna 501 and/or the second antenna 502. Although an example of a radio frequency system with two antennas is shown, a radio frequency system can include more or fewer antennas.

The front end systems 10, 20, 511, 514 of FIGS. 13A-13D can incorporate any of the front end systems described herein, such as those described with respect to FIGS. 1A-4, or any of the corresponding componentry such as any of the output matching networks 9 of those front end systems.

Multiple antennas can be included in a radio frequency system for a wide variety of reasons. In one example, the first antenna 501 and the second antenna 502 correspond to a transmit antenna and a receive antenna, respectively. In a second example, the first antenna 501 and the second antenna 502 are used for transmitting and/or receiving signals associated with different frequency ranges (for instance, different bands). In a third example, the first antenna 501 and the second antenna 502 support diversity communications, such as multiple-input multiple-output (MIMO) communications and/or switched diversity communications. In a fourth example, the first antenna 501 and the second antenna 502 support beamforming of transmit and/or receive signal beams.

Wireless Communication Devices

FIG. 14A is a schematic diagram of one example of a wireless communication device 650. The wireless communication device 650 includes a first antenna 641, a wireless personal area network (WPAN) system 651, a transceiver 652, a processor 653, a memory 654, a power management block 655, a second antenna 656, and a front end system 657.

Any of the suitable combination of features disclosed herein can be implemented in the wireless communication device 650. For example, the WPAN system 651 and/or the front end system 657 can be implemented using any of the features described above and/or in the sections below.

The WPAN system 651 is a front end system configured for processing radio frequency signals associated with personal area networks (PANs). The WPAN system 651 can be configured to transmit and receive signals associated with one or more WPAN communication standards, such as signals associated with one or more of LTE-M (LTE Machine Type Communication), Bluetooth, ZigBee, Z-Wave, Wireless USB, INSTEON, IrDA, or Body Area Network. In another embodiment, a wireless communication device can include a wireless local area network (WLAN) system in place of the illustrated WPAN system, and the WLAN system can process Wi-Fi signals.

FIG. 14B is a schematic diagram of another example of a wireless communication device 660. The illustrated wireless communication device 660 of FIG. 14B is a device configured to communicate over a PAN. This wireless communication device 660 can be relatively less complex than the wireless communication device 650 of FIG. 8A. As illustrated, the wireless communication device 660 includes an antenna 641, a WPAN system 651, a transceiver 662, a processor 653, and a memory 654. The WPAN system 660 can include any suitable combination of features disclosed herein. For example, the WPAN system 651 can be implemented using any of the features described above and/or in the sections below.

FIG. 14C is a schematic diagram of another example of a wireless communication device 800. The wireless communication device 800 includes a baseband system 801, a transceiver 802, a front-end system 803, one or more antennas 804, a power management system 805, a memory 806, a user interface 807, and a battery 808.

The wireless communication device 800 can be used to communicate using a wide variety of communications technologies, including, but not limited to, 2G, 3G, 4G (including LTE, LTE-Advanced, and LTE-Advanced Pro), 5G, WLAN (for instance, Wi-Fi), WPAN (for instance, LTE-M, Bluetooth and ZigBee), WMAN (for instance, WiMAX), and/or GPS technologies.

The transceiver 802 generates RF signals for transmission and processes incoming RF signals received from the antennas 804. It will be understood that various functionalities associated with the transmission and receiving of RF signals can be achieved by one or more components that are collectively represented in FIG. 14C as the transceiver 802. In one example, separate components (for instance, separate circuits or dies) can be provided for handling certain types of RF signals.

The front-end system 803 aids in conditioning signals transmitted to and/or received from the antennas 804. In the illustrated embodiment, the front-end system 803 includes one or more power amplifiers (PAs) 811, one or more low noise amplifiers (LNAs) 812, one or more filters 813, one or more switches 814, and one or more duplexers 815, and one

or more output matching networks 816. However, other implementations are possible.

For example, the front-end system 803 can provide a number of functionalities, including, but not limited to, amplifying signals for transmission, amplifying received signals, filtering signals, switching between different bands, switching between different power modes, switching between transmission and receiving modes, duplexing of signals, multiplexing of signals (for instance, duplexing or triplexing), or some combination thereof.

Any of the suitable combination of features disclosed herein can be implemented in the wireless communication device 800. For example, the front end system 803 can be implemented using any of the features described above and/or in the sections below.

In certain implementations, the wireless communication device 800 supports carrier aggregation, thereby providing flexibility to increase peak data rates. Carrier aggregation can be used for both Frequency Division Duplexing (FDD) and Time Division Duplexing (TDD), and may be used to aggregate a plurality of carriers or channels. Carrier aggregation includes contiguous aggregation, in which contiguous carriers within the same operating frequency band are aggregated. Carrier aggregation can also be non-contiguous, and can include carriers separated in frequency within a common band or in different bands.

The antennas 804 can include antennas used for a wide variety of types of communications. For example, the antennas 804 can include antennas for transmitting and/or receiving signals associated with a wide variety of frequencies and communications standards.

In certain implementations, the antennas 804 support MIMO communications and/or switched diversity communications. For example, MIMO communications use multiple antennas for communicating multiple data streams over a single radio frequency channel. MIMO communications benefit from higher signal to noise ratio, improved coding, and/or reduced signal interference due to spatial multiplexing differences of the radio environment. Switched diversity refers to communications in which a particular antenna is selected for operation at a particular time. For example, a switch can be used to select a particular antenna from a group of antennas based on a variety of factors, such as an observed bit error rate and/or a signal strength indicator.

The wireless communication device 800 can operate with beamforming in certain implementations. For example, the front-end system 803 can include phase shifters having variable phase controlled by the transceiver 802. Additionally, the phase shifters are controlled to provide beam formation and directivity for transmission and/or reception of signals using the antennas 804. For example, in the context of signal transmission, the phases of the transmit signals provided to the antennas 804 are controlled such that radiated signals from the antennas 804 combine using constructive and destructive interference to generate an aggregate transmit signal exhibiting beam-like qualities with more signal strength propagating in a given direction. In the context of signal reception, the phases are controlled such that more signal energy is received when the signal is arriving to the antennas 804 from a particular direction. In certain implementations, the antennas 804 include one or more arrays of antenna elements to enhance beamforming.

The baseband system 801 is coupled to the user interface 807 to facilitate processing of various user input and output (I/O), such as voice and data. The baseband system 801 provides the transceiver 802 with digital representations of transmit signals, which the transceiver 802 processes to

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generate RF signals for transmission. The baseband system **801** also processes digital representations of received signals provided by the transceiver **802**. As shown in FIG. **14C**, the baseband system **801** is coupled to the memory **806** of facilitate operation of the wireless communication device **800**.

The memory **806** can be used for a wide variety of purposes, such as storing data and/or instructions to facilitate the operation of the wireless communication device **800** and/or to provide storage of user information.

The power management system **805** provides a number of power management functions of the wireless communication device **800**. In certain implementations, the power management system **805** includes a PA supply control circuit that controls the supply voltages of the power amplifiers **811**. For example, the power management system **805** can be configured to change the supply voltage(s) provided to one or more of the power amplifiers **811** to improve efficiency, such as power added efficiency (PAE).

As shown in FIG. **14C**, the power management system **805** receives a battery voltage from the battery **808**. The battery **808** can be any suitable battery for use in the wireless communication device **800**, including, for example, a lithium-ion battery.

Some or all of the front end systems **656**, **803** or WPAN systems **651** of FIGS. **14A-14C** can incorporate any of the front end systems described herein, such as those described with respect to FIGS. **1A-4**, or any of the corresponding componentry such as any of the output matching networks **9** of those front end systems.

FIG. **15** is a schematic diagram of an example RF module **2010A** that includes a system-on-chip **2012A**, an RF front end IC **2012B**, a crystal **2012C**, and an integrated antenna **2014** according to an embodiment. The system-on-chip **2012A** can include one or more of a transceiver, processor (e.g., a baseband processor), and memory. The RF front end IC **2012B** can include any of the front end components described herein, such as any of the front end systems of FIGS. **1A-4**. For example, the RF front end IC **2012B** can include for example any of the power amplifiers, switches, low noise amplifiers, and output matching networks described herein. In some embodiments, the RF front end IC **2012B** is a semiconductor-on-insulator (e.g., silicon-on-insulator) die implementing FET-based power amplifier such as the cascade power amplifier of FIG. **3**. The RF module **2010A** can be a system in a package.

FIG. **15** shows the RF module **2010A** in plan view without a top shielding layer, which can also be included. As illustrated, the RF module **2010A** includes the component **2012A-2012C** on a package substrate **2016**, the antenna **2014** on the package substrate **2016**, and wire bonds **2018** attached to the package substrate **2016** and surrounding the components **2012A-2012C**. The antenna **2014** of the RF module **2010A** is outside of an RF shielding structure around the components **2012A-2012C**. Accordingly, the antenna **2014** can wirelessly receive and/or transmit RF signals without being shielded by the shielding structure around the components **2012A-2012C**. At the same time, the shielding structure can provide RF isolation between the components **2012A-2012C** and the antenna **2014** and/or other electronic components.

The components **2012A-2012C** can include any suitable circuitry configured to receive, process, and/or provide an RF signal. In certain implementations, the RF front end IC **2012B** can include a power amplifier, a low-noise amplifier, an RF switch, a filter, a matching network, or any combination thereof, and can be clocked by a signal derived from

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the crystal **2012C**. An RF signal can have a frequency in the range from about 30 kHz to 300 GHz. In accordance with certain communications standards, an RF signal can be in a range from about 450 MHz to about 6 GHz, in a range from about 700 MHz to about 2.5 GHz, or in a range from about 2.4 GHz to about 2.5 GHz. In certain implementations, the RF component **2012** can receive and/or provide signals in accordance with a wireless personal area network (WPAN) standard, such as LTE-M, Bluetooth, ZigBee, Z-Wave, Wireless USB, INSTEON, IrDA, or Body Area Network. In some other implementations, the RF component can receive and/or provide signals in accordance with a wireless local area network (WLAN) standard, such as Wi-Fi.

The antenna **2014** can be any suitable antenna configured to receive and/or transmit RF signals. The antenna **2014** can be a folded monopole antenna in certain applications. The antenna **2014** can be any suitable shape. For instance, the antenna **2014** can have a meandering shape as shown in FIG. **15**. In other embodiments, the antenna can be U-shaped, coil shaped, or any other suitable shape for a particular application. The antenna **2014** can transmit and/or receive RF signals associated with the RF front end IC **2012B**. The antenna **2014** can occupy any suitable amount of area of the packaging substrate **2016**. For instance, the antenna **2014** can occupy from about 10% to 75% of the area of the package substrate **2016** in certain implementations.

The antenna **2014** can be printed on the packaging substrate **2016**. A printed antenna can be formed from one or more conductive traces on the packaging substrate **2016**. The one or more conductive traces can be formed by etching a metal pattern on the packaging substrate **2016**. A printed antenna can be a microstrip antenna. Printed antennas can be manufactured relatively inexpensively and compactly due to, for example, their 2-dimensional physical geometries. Printed antennas can have a relatively high mechanical durability.

The package substrate **2016** can be a laminate substrate. The package substrate **2016** can include one or more routing layers, one or more insulating layers, a ground plane, or any combination thereof. In certain applications, the package substrate can include four layers. The RF front end IC **2012B** can be electrically connected to the antenna **2014** by way of metal routing in a routing layer of the packaging substrate **2016** in certain applications.

The wire bonds **2018** are part of an RF shielding structure around the RF component **2012**. An RF shielding structure can be any shielding structure configured to provide suitable shielding associated with RF signals. The wire bonds **2018** can provide RF isolation between the antenna **2014** and some or all of the components **2012A-2012C** so as to prevent electromagnetic interference between these components from significantly impacting performance of the antenna **2014** and/or some or all of the components **2012A-2012C**. The wire bonds **2018** can surround the RF component **2012** as illustrated. The wire bonds **2018** can be arranged around the components **2012A-2012C** in any suitable arrangement, which can be rectangular as illustrated or non-rectangular in some other implementations. In the RF module **2010A** illustrated in FIG. **15**, the wire bonds **2018** form four walls around the components **2012A-2012C**. The wire bonds **2018** can be arranged such that adjacent wire bonds are spaced apart from each other by a distance to provide sufficient RF isolation between the components **2012A-2012C** and other electronic components.

FIG. **15** illustrates an RF module in accordance with the principles and advantages discussed herein. The RF module can be selectively shielded, where various RF components

can be implemented within a shielding structure. For instance, FIG. 15 shows an example of an RF component that includes three different elements. Other RF components can alternatively or additionally be implemented. A conformal layer can be disposed along at least one side the RF component of the RF module in embodiments in which a shielding layer is formed after singulation of the RF modules. The conformal structure can include any suitable conductive material. For example, the conductive conformal structure can include the same conductive material as the shielding layer in certain applications

Any of the embodiments described above can be implemented in association with mobile devices such as cellular handsets. The principles and advantages of the embodiments can be used for any systems or apparatus, such as any uplink wireless communication device, that could benefit from any of the embodiments described herein. The teachings herein are applicable to a variety of systems. Although this disclosure includes example embodiments, the teachings described herein can be applied to a variety of structures. Any of the principles and advantages discussed herein can be implemented in association with RF circuits configured to process signals having a frequency in a range from about 30 kHz to 300 GHz, such as in a frequency range from about 400 MHz to 25 GHz.

Aspects of this disclosure can be implemented in various electronic devices. Examples of the electronic devices can include, but are not limited to, consumer electronic products, parts of the consumer electronic products such as packaged radio frequency modules, radio frequency filter die, uplink wireless communication devices, wireless communication infrastructure, electronic test equipment, etc. Examples of the electronic devices can include, but are not limited to, a mobile phone such as a smart phone, a wearable computing device such as a smart watch or an ear piece, a telephone, a television, a computer monitor, a computer, a modem, a hand-held computer, a laptop computer, a tablet computer, a microwave, a refrigerator, a vehicular electronics system such as an automotive electronics system, a robot such as an industrial robot, an Internet of things device, a stereo system, a digital music player, a radio, a camera such as a digital camera, a portable memory chip, a home appliance such as a washer or a dryer, a peripheral device, a wrist watch, a clock, etc. Further, the electronic devices can include unfinished products.

Unless the context indicates otherwise, throughout the description and the claims, the words “comprise,” “comprising,” “include,” “including” and the like are to generally be construed in an inclusive sense, as opposed to an exclusive or exhaustive sense; that is to say, in the sense of “including, but not limited to.” Conditional language used herein, such as, among others, “can,” “could,” “might,” “may,” “e.g.,” “for example,” “such as” and the like, unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements and/or states. The word “coupled”, as generally used herein, refers to two or more elements that may be either directly coupled, or coupled by way of one or more intermediate elements. Likewise, the word “connected”, as generally used herein, refers to two or more elements that may be either directly connected, or connected by way of one or more intermediate elements. Additionally, the words “herein,” “above,” “below,” and words of similar import, when used in this application, shall refer to this application as a whole and not to any particular portions of this application. Where the

context permits, words in the above Detailed Description using the singular or plural number may also include the plural or singular number respectively.

While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the disclosure. Indeed, the novel filters, multiplexer, devices, modules, wireless communication devices, apparatus, methods, and systems described herein may be embodied in a variety of other forms. Furthermore, various omissions, substitutions and changes in the form of the filters, multiplexer, devices, modules, wireless communication devices, apparatus, methods, and systems described herein may be made without departing from the spirit of the disclosure. For example, while blocks are presented in a given arrangement, alternative embodiments may perform similar functionalities with different components and/or circuit topologies, and some blocks may be deleted, moved, added, subdivided, combined, and/or modified. Each of these blocks may be implemented in a variety of different ways. Any suitable combination of the elements and/or acts of the various embodiments described above can be combined to provide further embodiments. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosure.

What is claimed is:

1. A semiconductor-on-insulator die comprising:

a substrate layer, an active layer, an insulator layer between the substrate layer and the active layer, a first metal layer, and a second metal layer, the first metal layer between the active layer and the second metal layer;

at least one contact pad, the second metal layer between the at least one contact pad and the first metal layer;

a transistor including a first terminal, the first terminal formed within the active layer; and

a conductive path extending between the first terminal and the at least one contact pad through at least the first metal layer and the second metal layer, at least a first portion of the conductive path extending in a straight-line path from the first terminal to the at least one contact pad, the at least one contact pad including a first contact pad and a second contact pad, the first portion of the conductive path extending in a straight-line path from the first terminal to the first contact pad, and a second portion of the conductive path extending in a straight-line path from the first terminal to the second contact pad.

2. The semiconductor-on-insulator die of claim 1 wherein the transistor is a field effect transistor and the first terminal is a source of the transistor.

3. The semiconductor-on-insulator die of claim 1 wherein the conductive path further includes at least a second portion extending between the first terminal and the at least one contact pad and which resides at least partly outside of a footprint of the at least one contact pad.

4. The semiconductor-on-insulator die of claim 1 wherein a footprint of the at least one contact pad overlaps with a footprint of the transistor.

5. The semiconductor-on-insulator die of claim 4 wherein at least 50 percent of the first terminal falls within the footprint of the at least one contact pad.

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6. The semiconductor-on-insulator die of claim 1 wherein the semiconductor-on-insulator die is a flip-chip die and the at least one contact pad includes at least one contact bump.

7. A semiconductor-on-insulator die comprising:

a substrate layer, an active layer, an insulator layer between the substrate layer and the active layer, a first metal layer, and a second metal layer, the first metal layer between the active layer and the second metal layer;

at least one contact pad, the second metal layer between the at least one contact pad and the first metal layer; a transistor including a first terminal, the first terminal formed within the active layer; and

a conductive path extending between the first terminal and the at least one contact pad through at least the first metal layer and the second metal layer, at least a first portion of the conductive path extending in a straight-line path from the first terminal to the at least one contact pad, the conductive path including a first via segment formed within a first via layer and a first metal segment formed within the first metal layer, the first via layer between the active layer and the first metal layer, a surface area of an interface between the first via segment and the first terminal being at least 5 percent of a surface area of the first terminal.

8. The semiconductor-on-insulator die of claim 7 wherein the first terminal substantially completely overlaps with the first via segment.

9. The semiconductor-on-insulator die of claim 7 further comprising a second via layer, the second via layer between the first metal layer and the second metal layer, the conductive path further including a second via segment formed from the second via layer and a second metal segment formed from the second metal layer.

10. The semiconductor-on-insulator die of claim 9 wherein the surface area of an interface between the first metal segment and the second via segment is at least 15 percent of the surface area of the first metal segment.

11. A semiconductor-on-insulator die comprising:

a substrate layer, an active layer, an insulator layer between the substrate layer and the active layer, a first metal layer, and a second metal layer, the first metal layer between the active layer and the second metal layer;

at least one contact pad, the second metal layer between the at least one contact pad and the first metal layer; a transistor including a first terminal, the first terminal formed within the active layer; and

a conductive path extending between the first terminal and the at least one contact pad through at least the first metal layer and the second metal layer, at least a first portion of the conductive path extending in a straight-line path from the first terminal to the at least one contact pad, the transistor including a first section and a second section, the at least one contact pad including a first contact pad and a second contact pad, a footprint of the first contact pad overlapping with a footprint of the first section, and a footprint of the second contact pad overlapping with a footprint of the second section.

12. The semiconductor-on-insulator die of claim 11 wherein the first portion of the conductive path extends in a straight-line path from the first terminal to the first contact pad, and a second portion of the conductive path extends in a straight-line path from the first terminal to the second contact pad.

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13. A packaged module comprising:

a semiconductor-on-insulator die including a substrate layer, an active layer, an insulator layer between the substrate layer and the active layer, a first metal layer, and a second metal layer, the first metal layer between the active layer and the second metal layer, the die further including at least one contact pad and a transistor including a first terminal, the first terminal formed within the active layer, the second metal layer between the at least one contact pad and the first metal layer;

a ground plane electrically connected to the at least one contact pad; and

a conductive path extending between the first terminal and the ground plane through at least the first metal layer, the second metal layer, and the at least one contact pad, at least a first portion of the conductive path extending in a straight-line path from the first terminal to the ground plane.

14. The packaged module of claim 13 wherein the at least one contact pad includes a first contact pad and a second contact pad, the first portion of the conductive path extends in a straight-line path from the first terminal to the ground plane through at least the first metal layer, the second metal layer, and the first contact pad, and a second portion of the conductive path extends in a straight-line path from the first terminal to the ground plane through at least the first metal layer, the second metal layer, and the second contact pad.

15. The packaged module of claim 13 further comprising a module metal layer between the ground plane and the at least one contact pad.

16. The packaged module of claim 15 further comprising a module via layer between the module metal layer and the ground plane, and a module via segment formed within the module via layer, the conductive path further including the module via segment.

17. A power amplifier system comprising:

a semiconductor-on-insulator die including a substrate layer, an active layer, an insulator layer between the substrate layer and the active layer, a first metal layer, and a second metal layer, the first metal layer between the active layer and the second metal layer, the die further including at least one contact pad and a transistor including a first terminal, the first terminal formed within the active layer, the second metal layer between the at least one contact pad and the first metal layer; and

a power amplifier configured to amplify a radio frequency signal and including a first transistor formed in the semiconductor-on-insulator die, the first transistor including a first terminal formed within the active layer of the semiconductor-on-insulator die, a conductive path extending between the first terminal and the at least one contact pad through at least the first metal layer and the second metal layer, at least a first portion of the conductive path extending in a straight-line path from the first terminal to the at least one contact pad.

18. The power amplifier system of claim 17 wherein the first transistor is a field effect transistor and the first terminal is a source terminal.

19. The power amplifier system of claim 18 wherein the power amplifier further includes a second transistor formed in the semiconductor-on-insulator die, the second transistor being a field effect transistor.

20. The power amplifier system of claim 17 wherein the at least one contact pad includes a first contact pad and a second contact pad, the first portion of the conductive path

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extends in a straight-line path from the first terminal to the first contact pad, and a second portion of the conductive path extends in a straight-line path from the first terminal to the second contact pad.

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